



US 20250287701A1

(19) **United States**(12) **Patent Application Publication**  
**Wang et al.**(10) **Pub. No.: US 2025/0287701 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **FORMING ESD DEVICES USING  
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**Ming-Shuan Li**, Hsinchu County (TW)(21) Appl. No.: **19/218,263**(22) Filed: **May 24, 2025****Related U.S. Application Data**

- (63) Continuation of application No. 18/623,294, filed on Apr. 1, 2024, now Pat. No. 12,317,602, which is a continuation of application No. 18/305,556, filed on Apr. 24, 2023, now Pat. No. 11,948,936, which is a continuation of application No. 17/224,671, filed on Apr. 7, 2021, now Pat. No. 11,637,099.
- (60) Provisional application No. 63/039,293, filed on Jun. 15, 2020.

**Publication Classification**

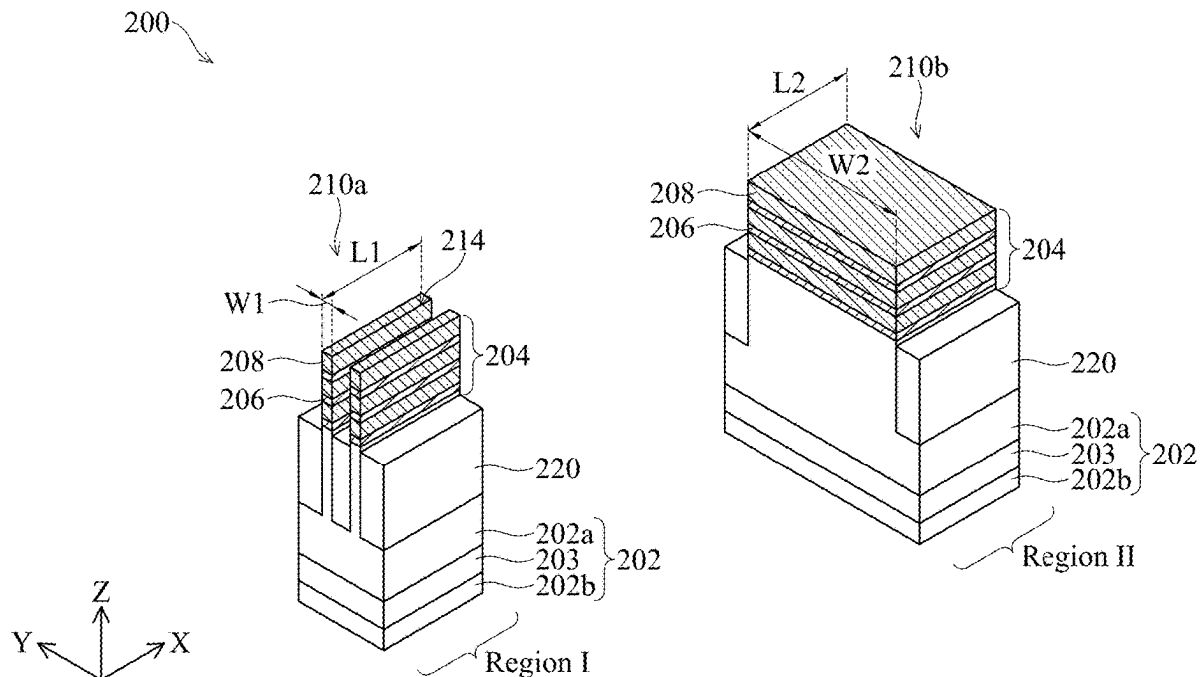
- (51) **Int. Cl.**  
**H10D 89/60** (2025.01)  
**H10D 30/01** (2025.01)  
**H10D 30/67** (2025.01)

**H10D 64/01** (2025.01)**H10D 84/01** (2025.01)**H10D 84/03** (2025.01)

(52) **U.S. Cl.**  
CPC ..... **H10D 89/811** (2025.01); **H10D 30/024**  
(2025.01); **H10D 30/6735** (2025.01); **H10D**  
**30/6757** (2025.01); **H10D 64/017** (2025.01);  
**H10D 84/0128** (2025.01); **H10D 84/0135**  
(2025.01); **H10D 84/0158** (2025.01); **H10D**  
**84/038** (2025.01); **H10D 89/931** (2025.01)

(57) **ABSTRACT**

The present disclosure provides a semiconductor device. The semiconductor device includes a stack disposed in a first region of the semiconductor device, a plurality of channel members disposed in a second region of the semiconductor device and vertically stacked, and a metal gate wrapping around at least one of the channel members. The stack has first type epitaxial layers and second type epitaxial layers alternately arranged in a vertical direction. The first type epitaxial layers have a first material composition and the second type epitaxial layers have a second material composition different from the first material composition. The stack has a first implant region of a first conductivity type and a second implant region of a second conductivity type opposite the first conductivity type. The channel members have the first material composition.



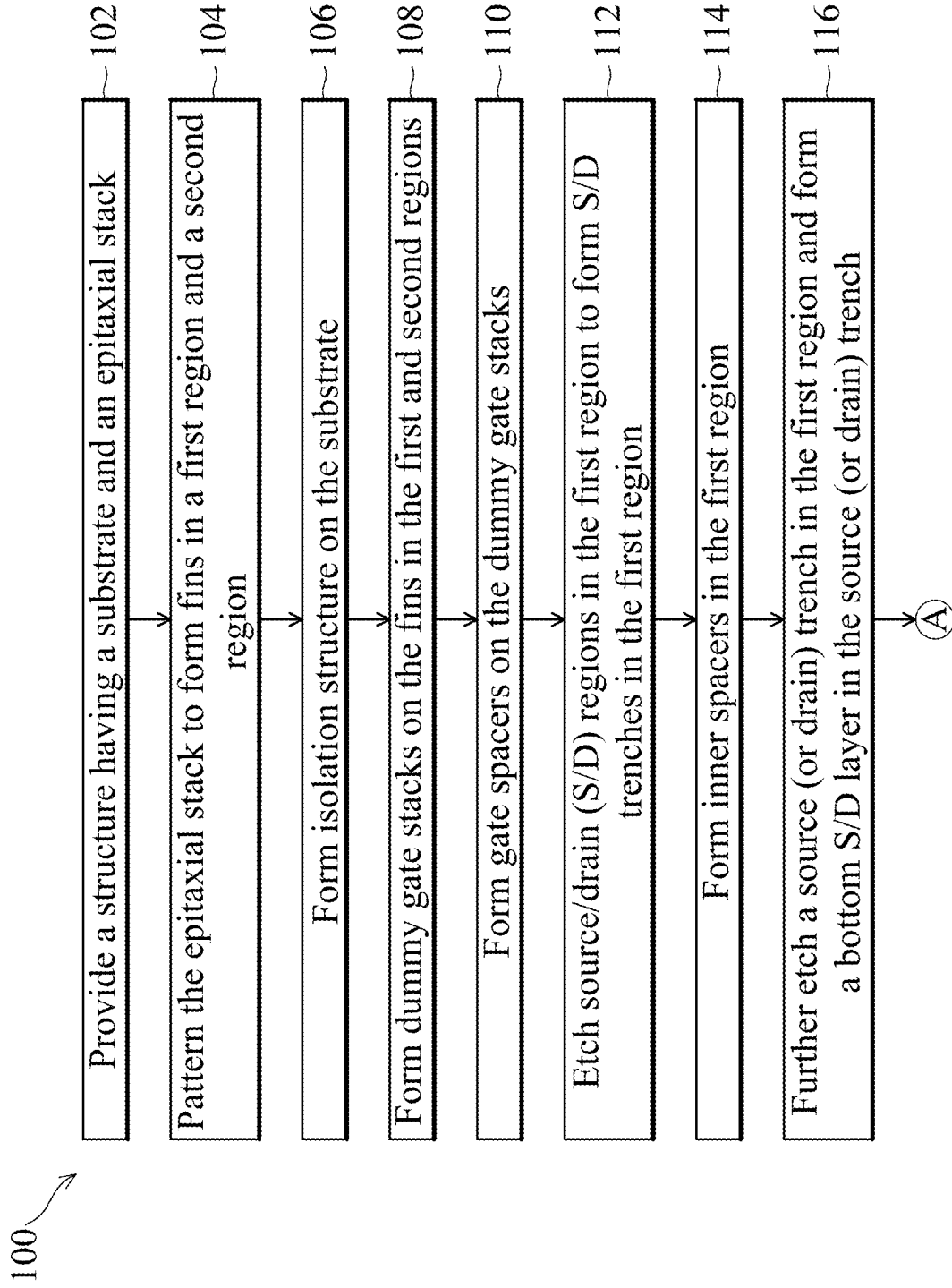


FIG. 1A

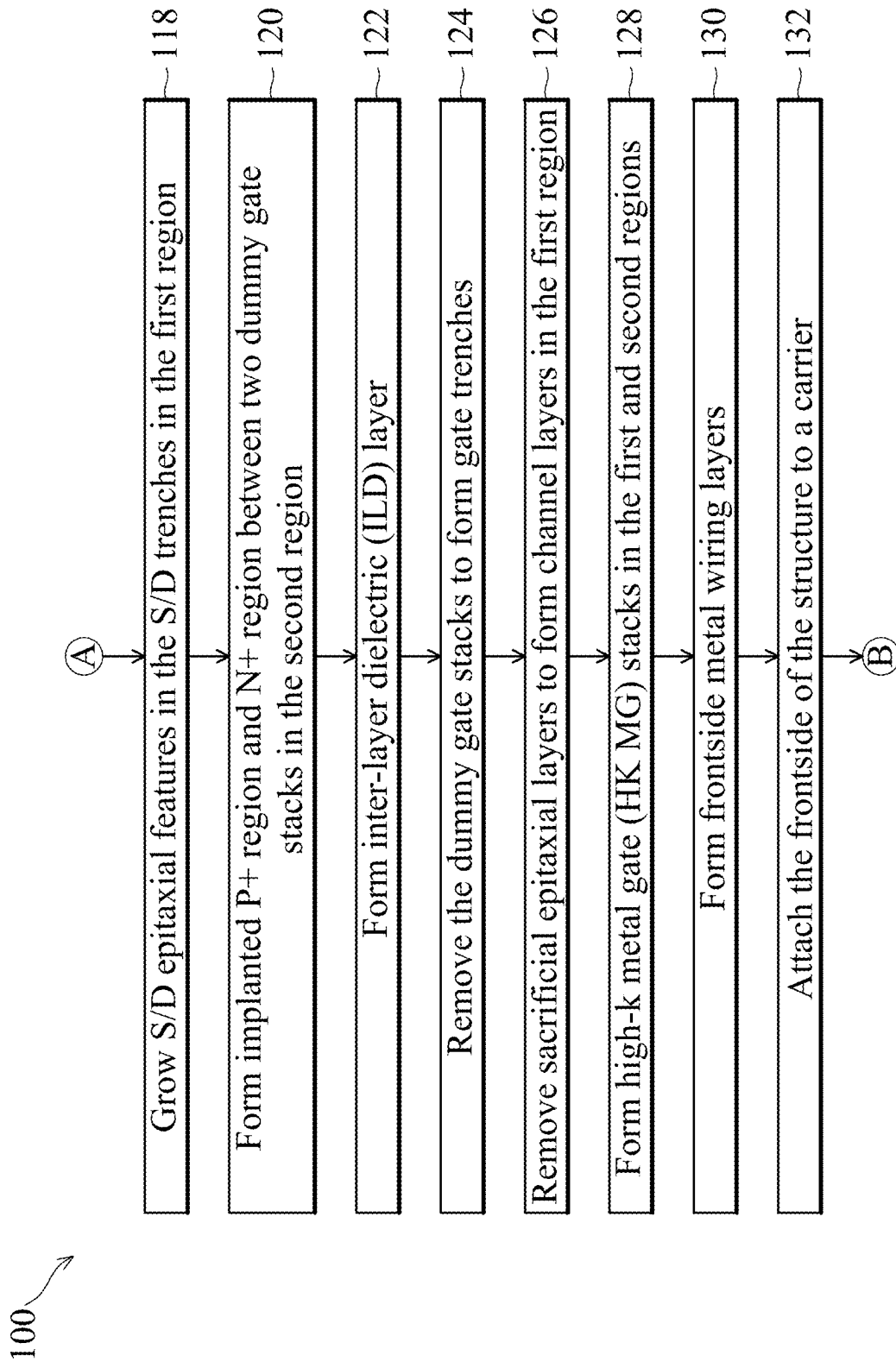


FIG. 1B

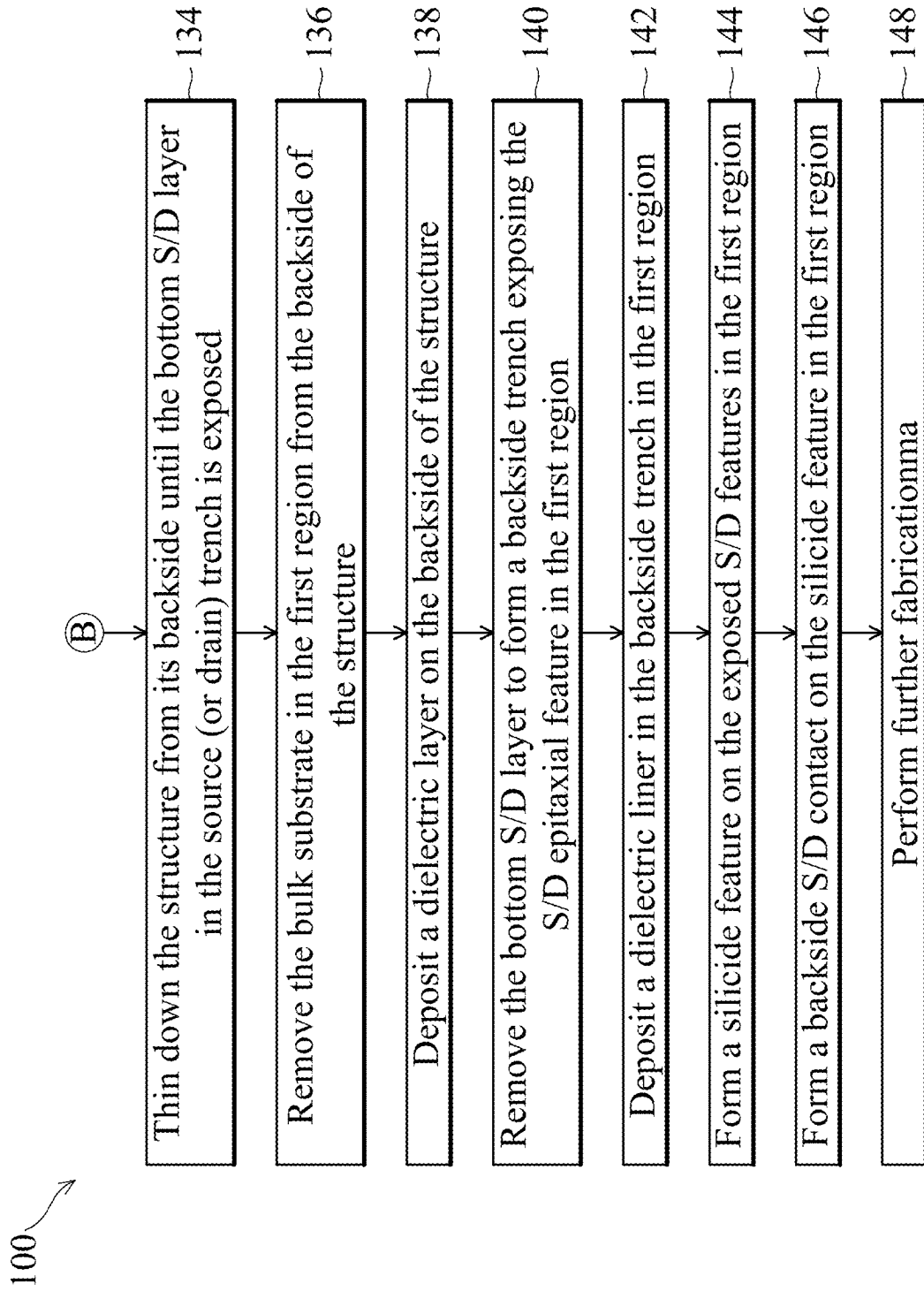


FIG. 1C

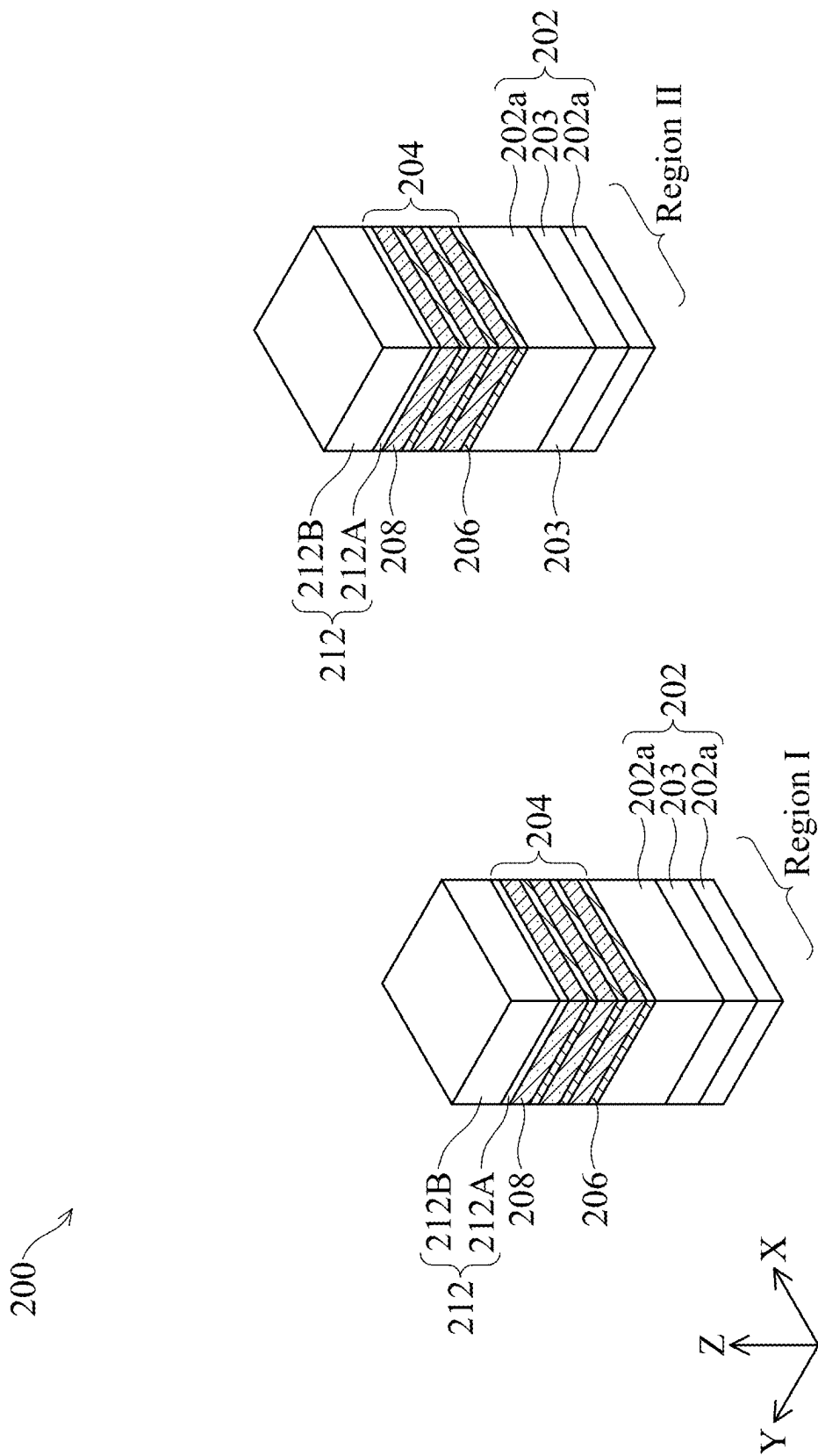


FIG. 2

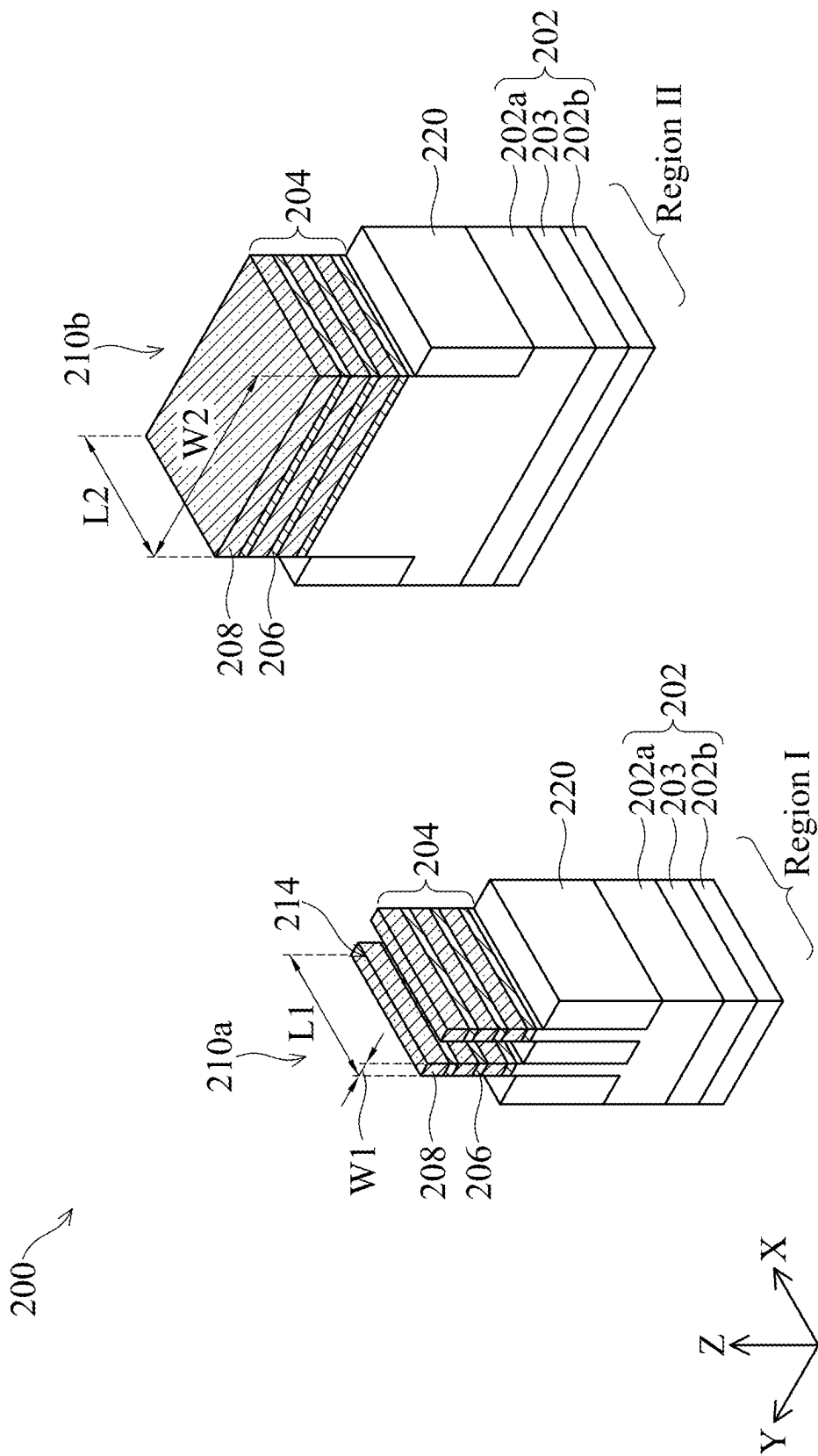
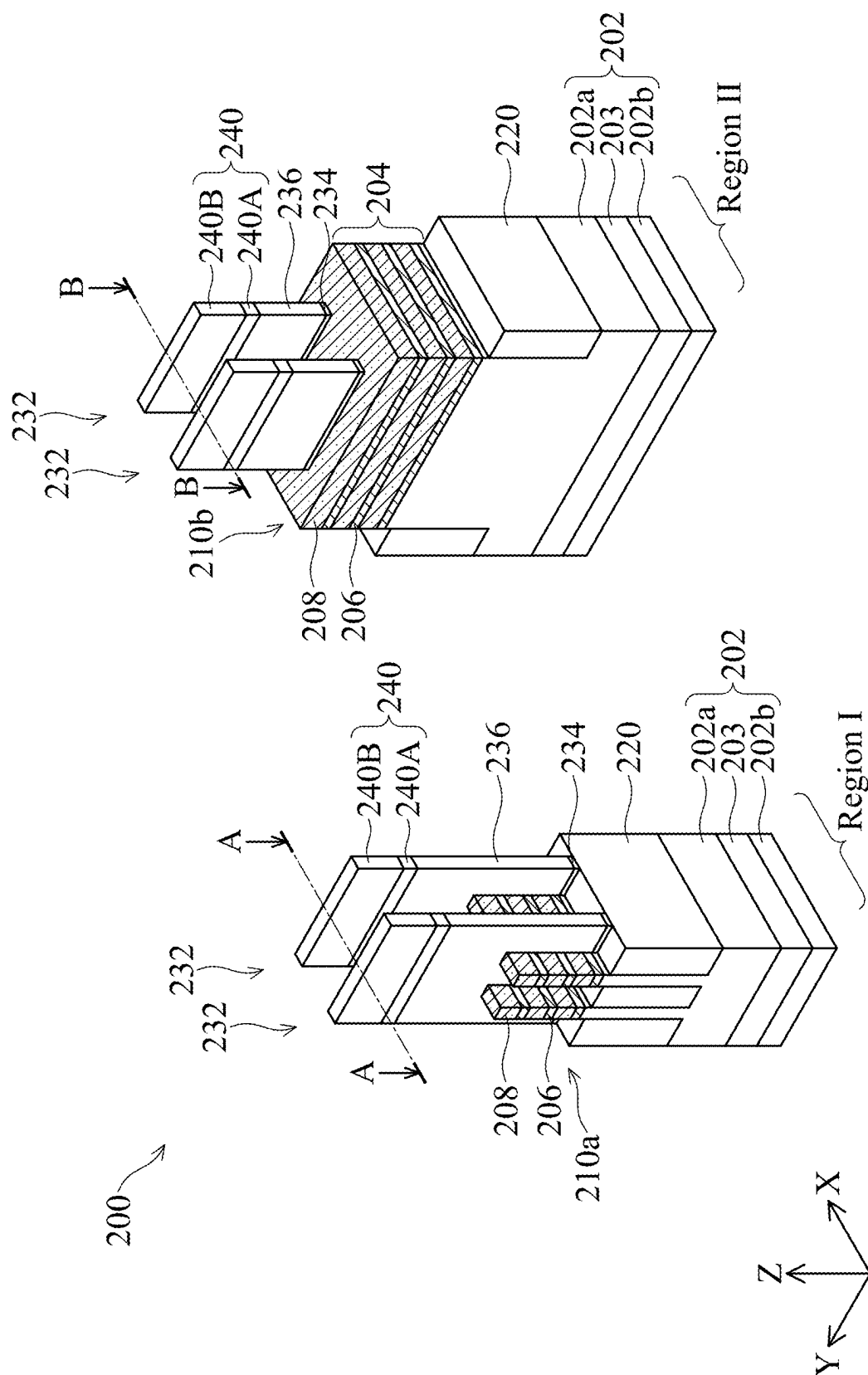


FIG. 3



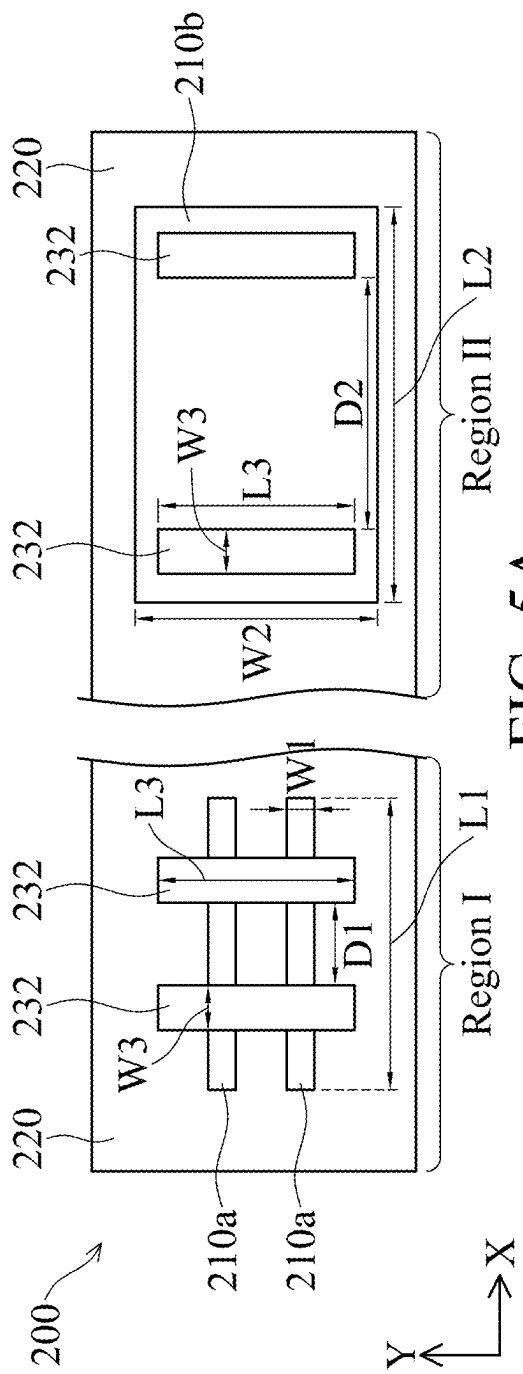


FIG. 5A

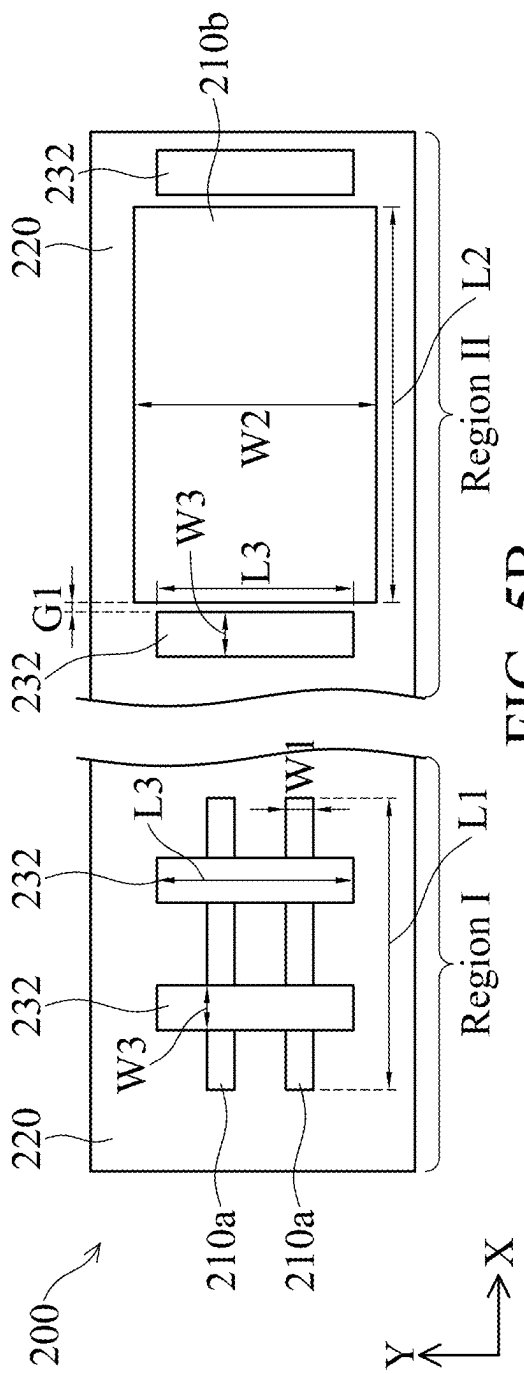


FIG. 5B



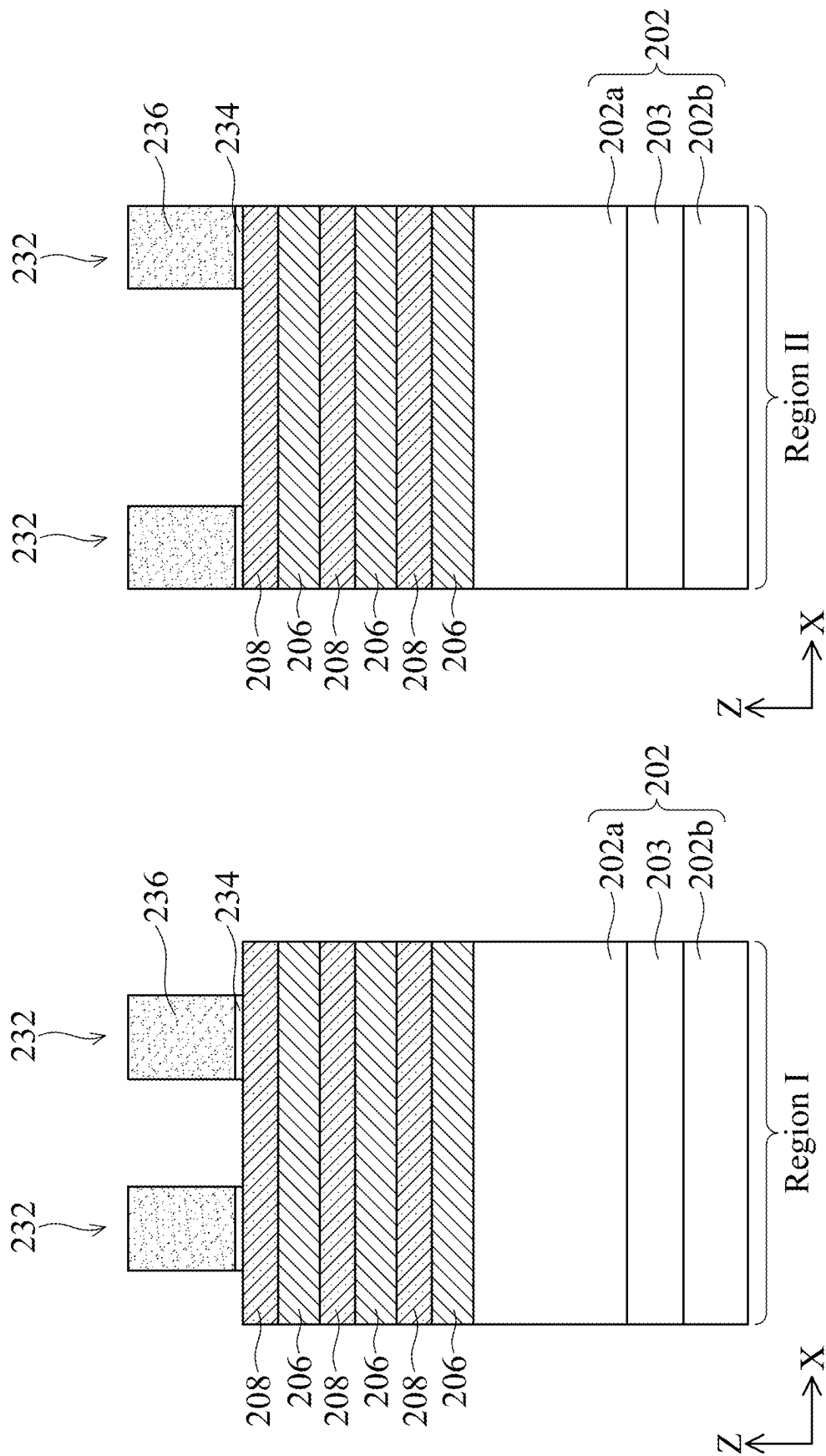


FIG. 6B

FIG. 6A

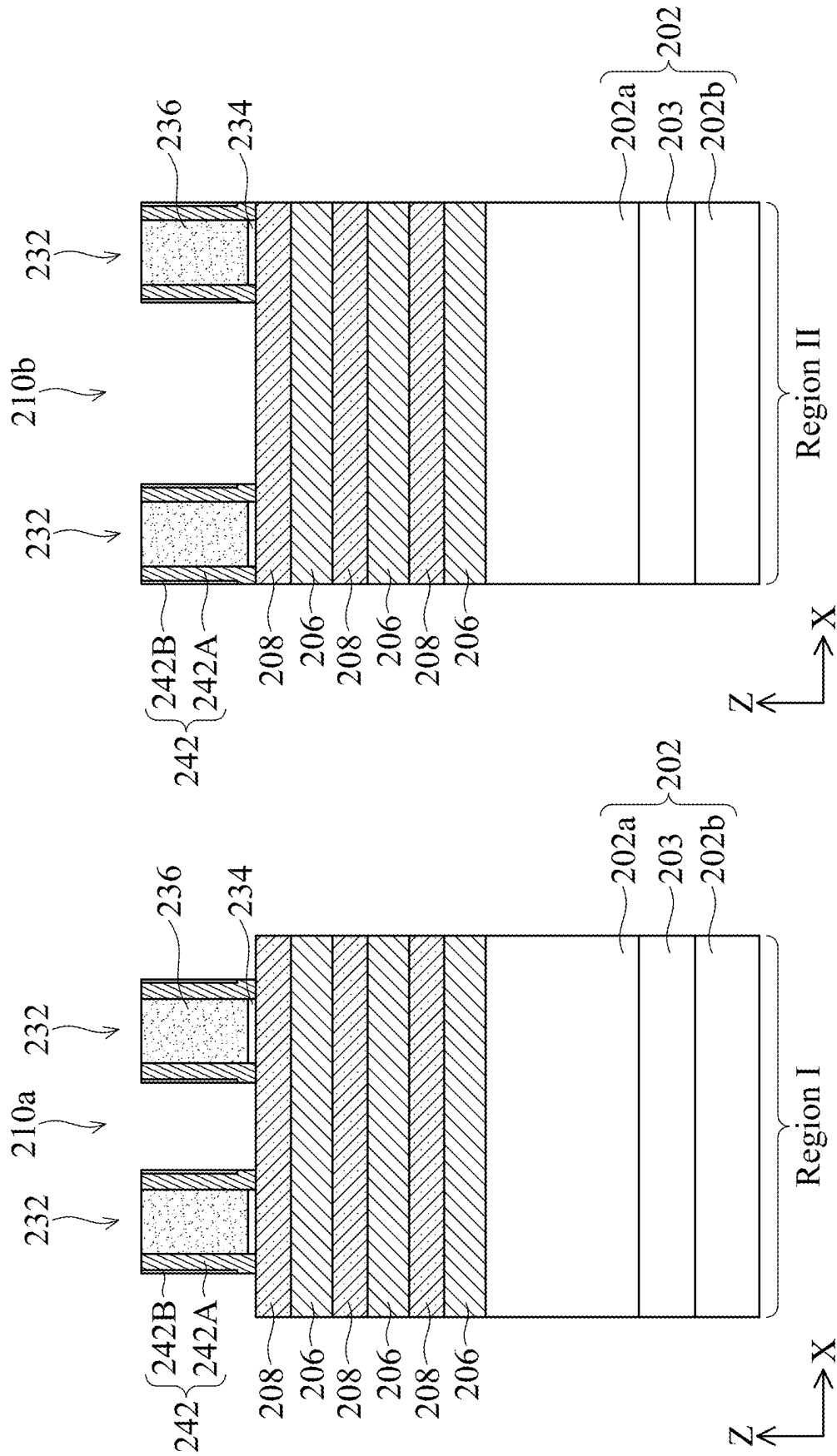


FIG. 7B

FIG. 7A

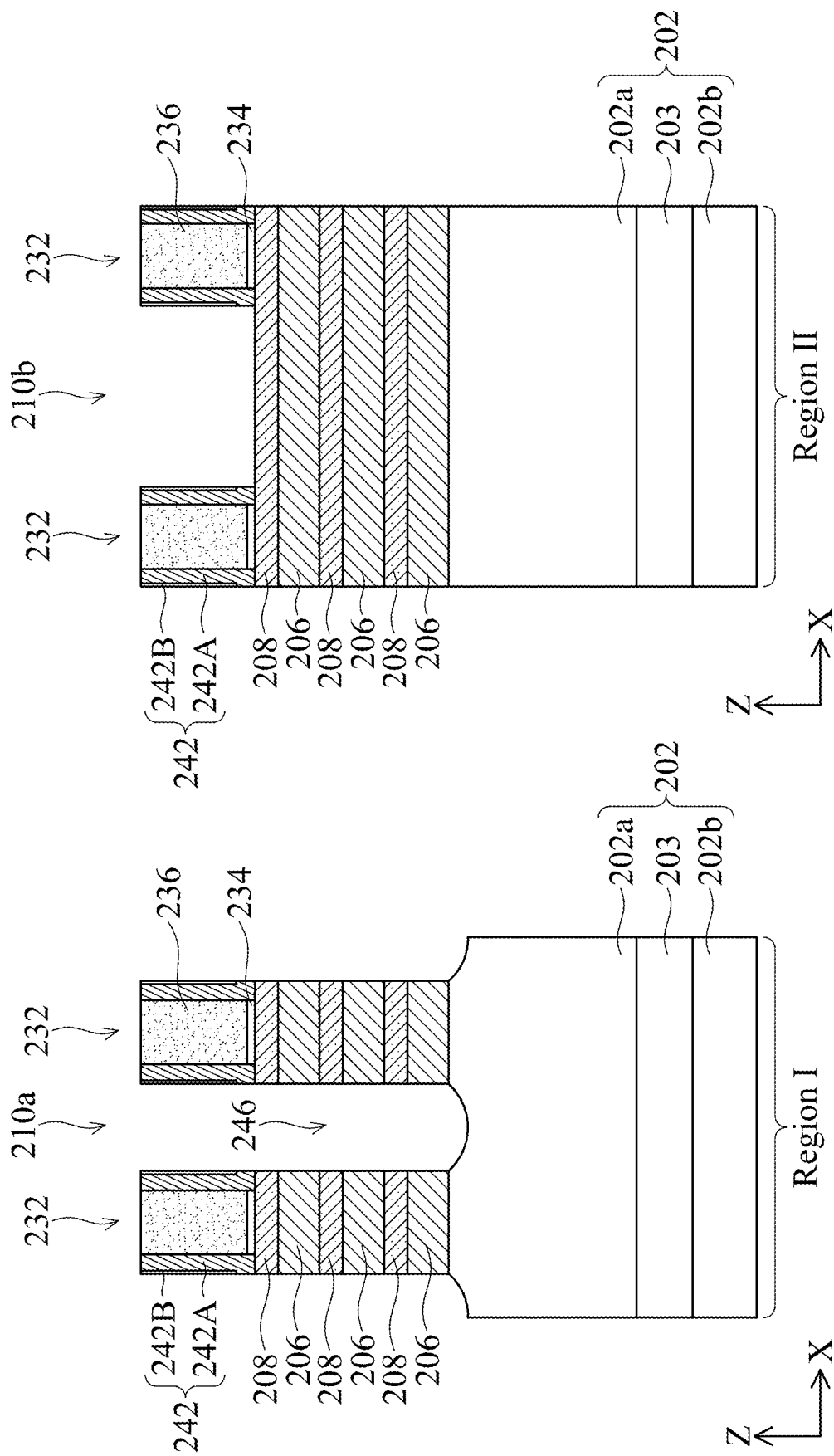
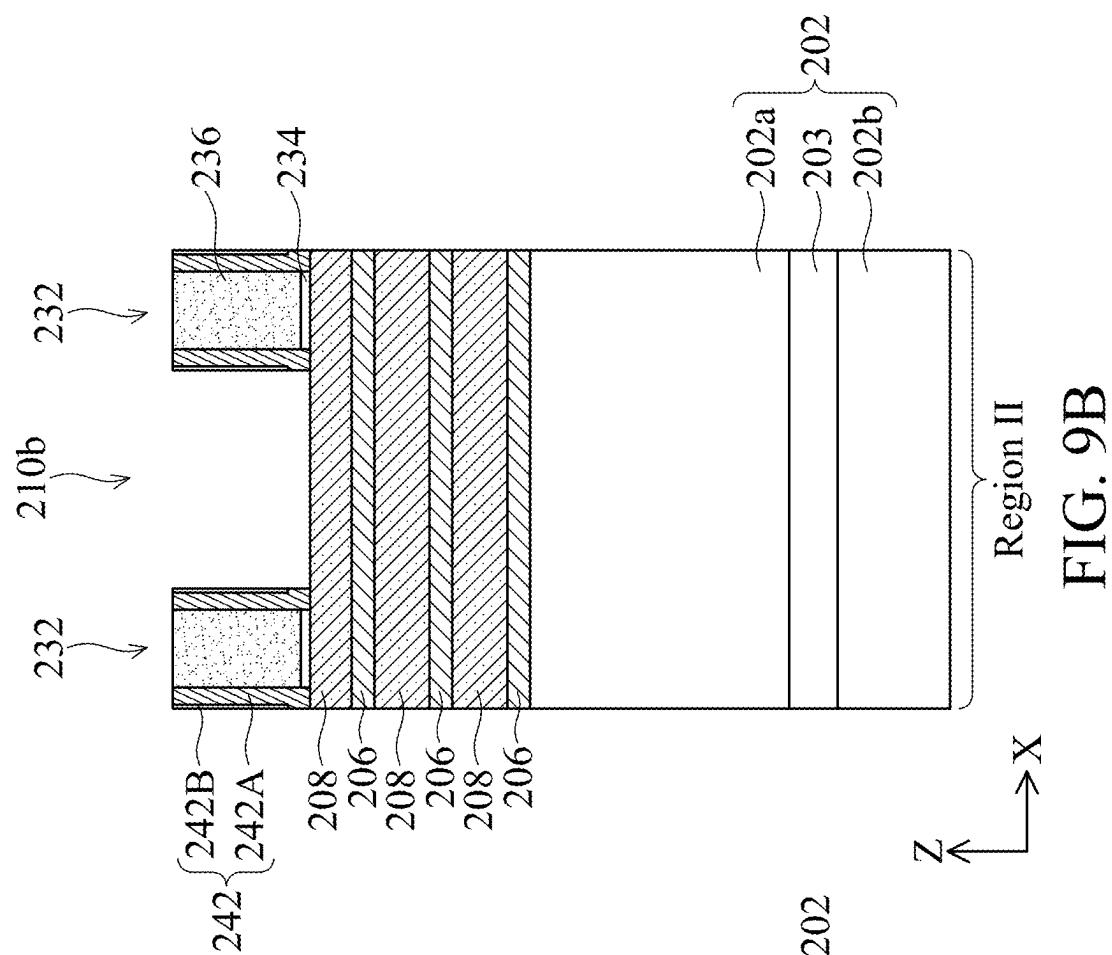
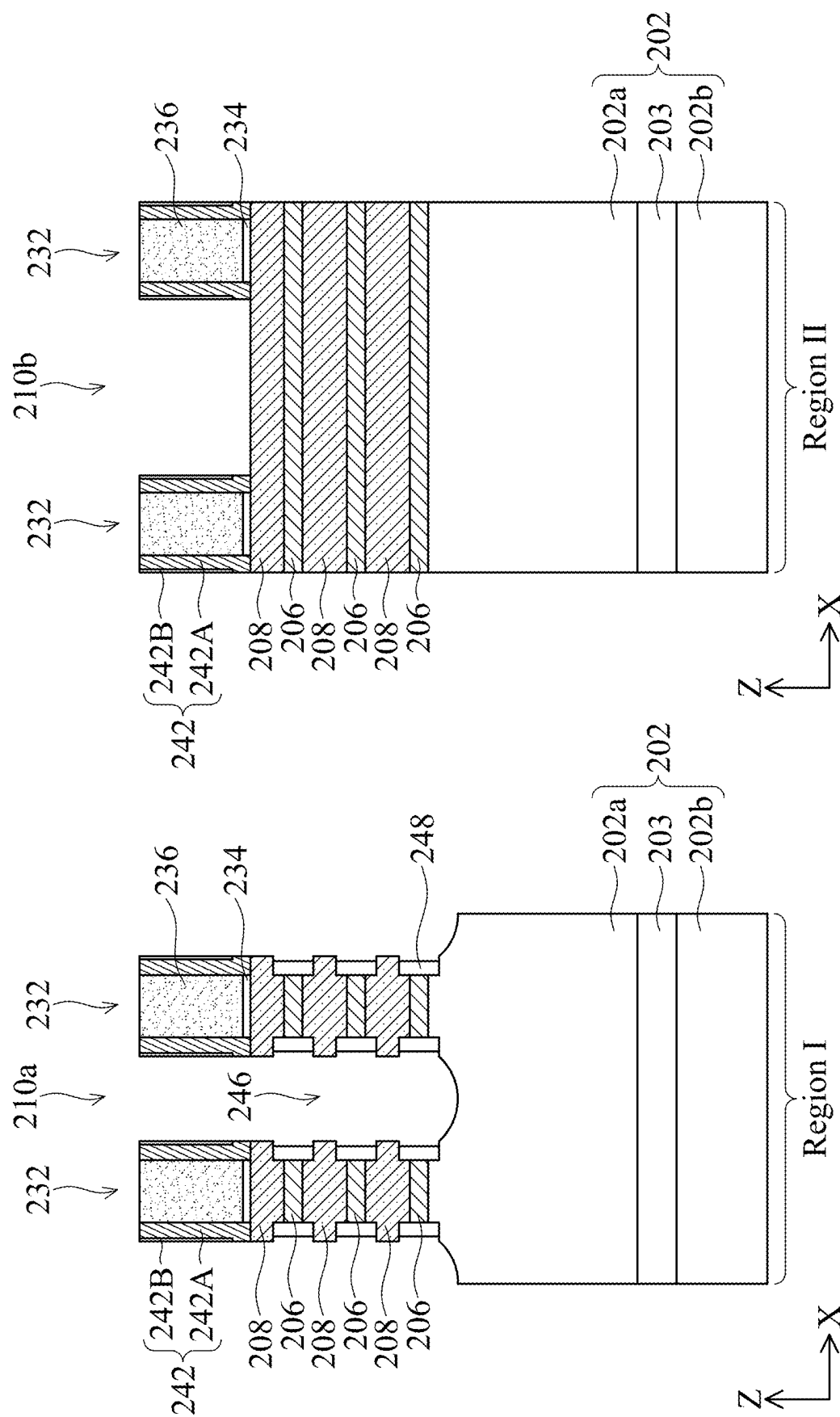


FIG. 8B

FIG. 8A



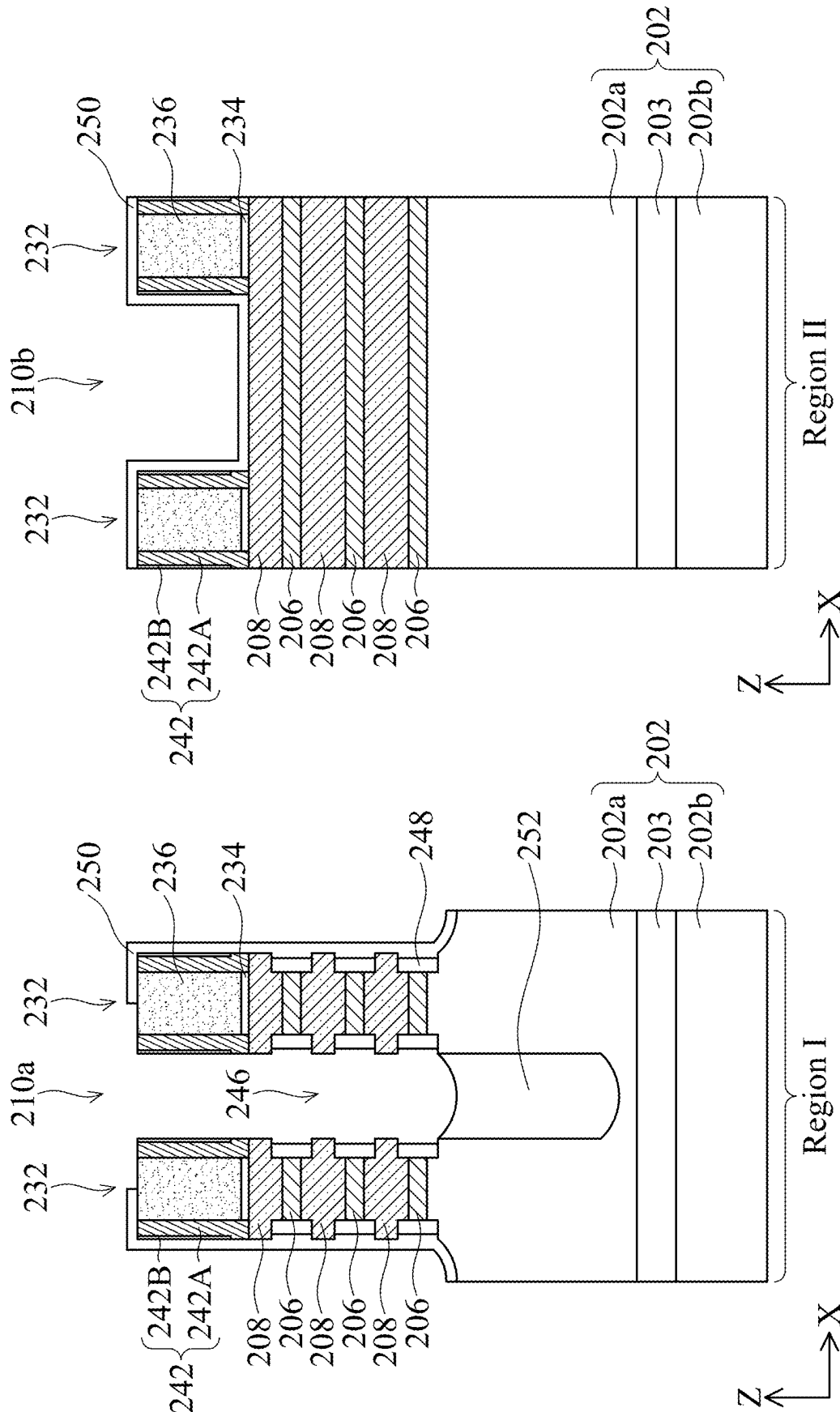


FIG. 10B

FIG. 10A

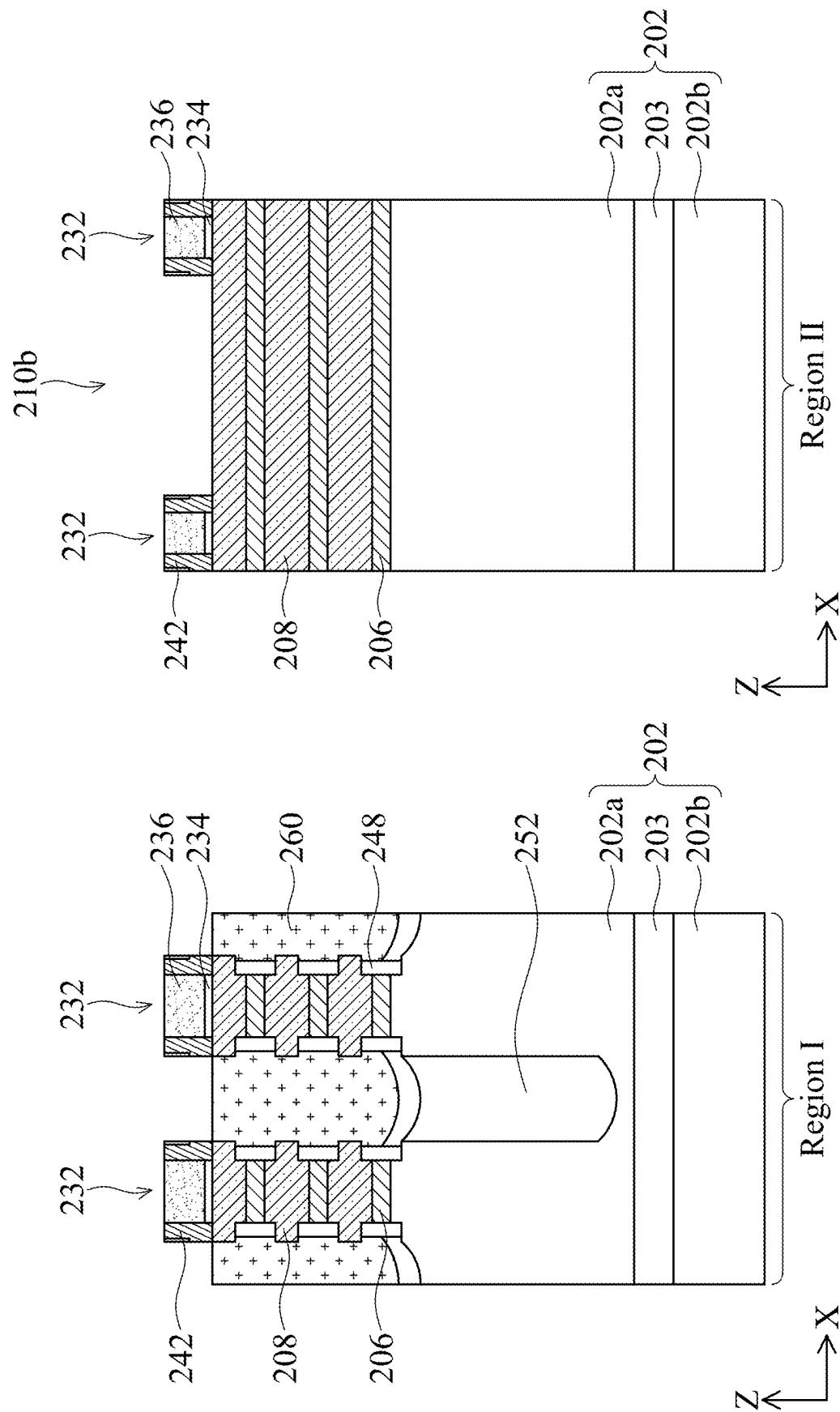
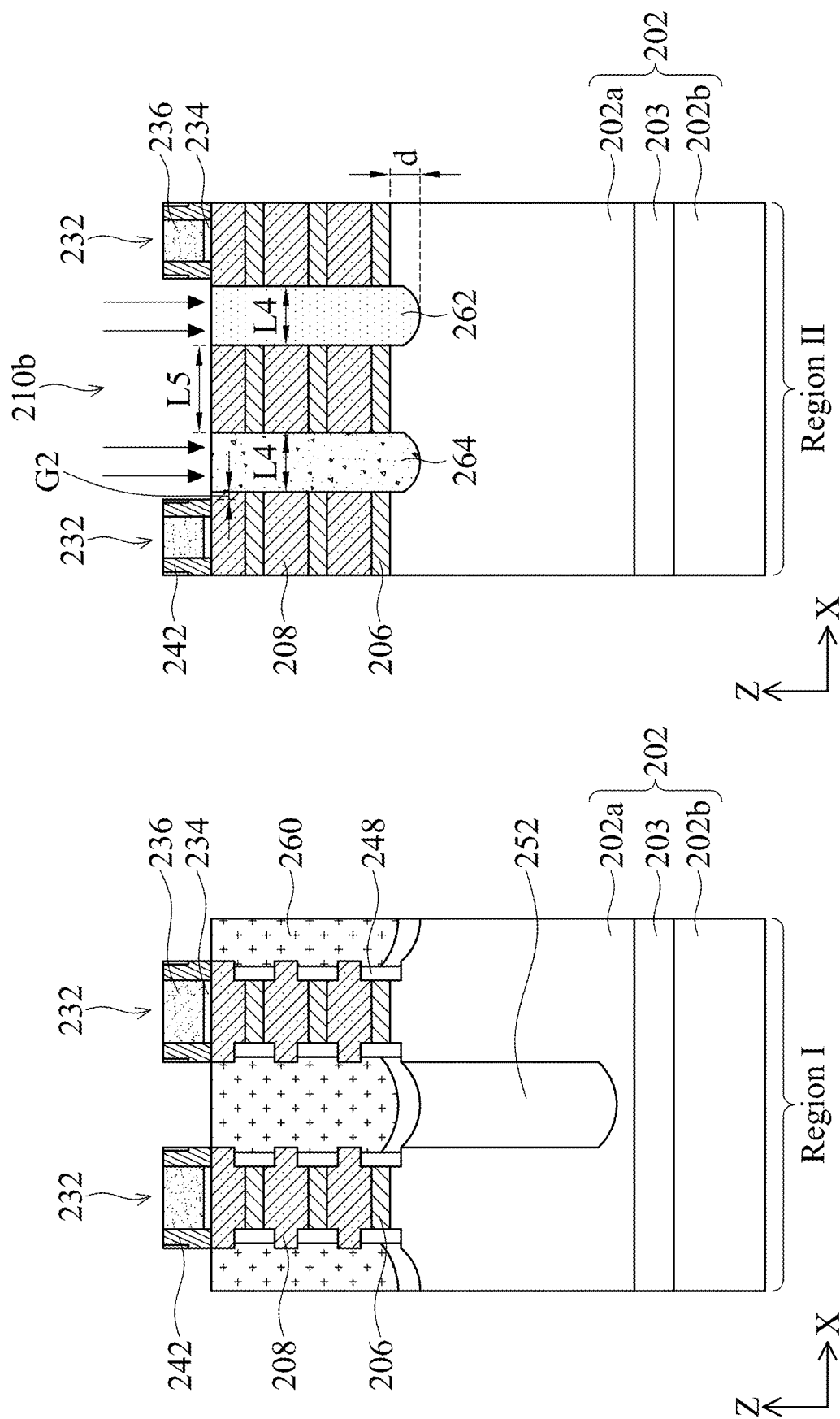


FIG. 11B

FIG. 11A



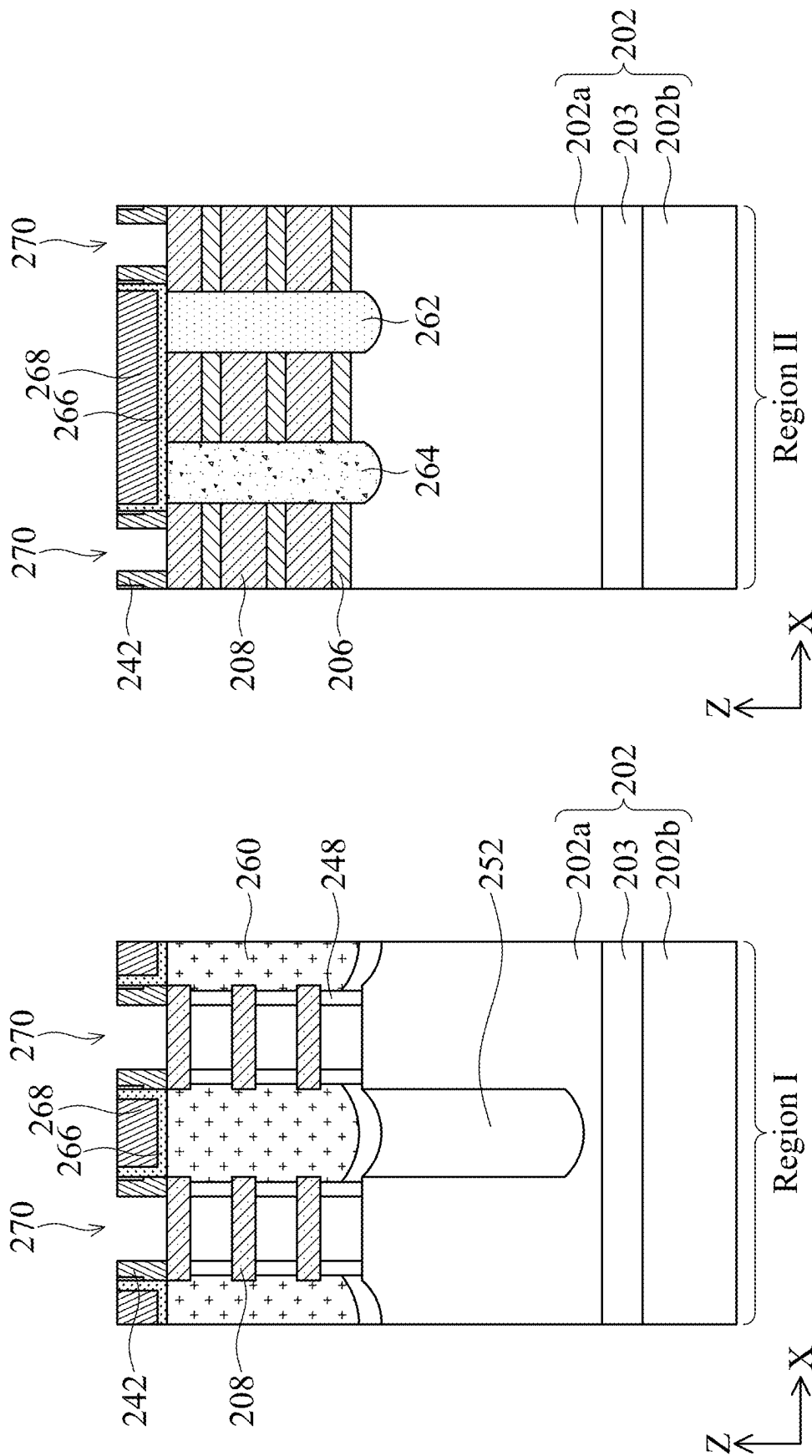


FIG. 13B

FIG. 13A



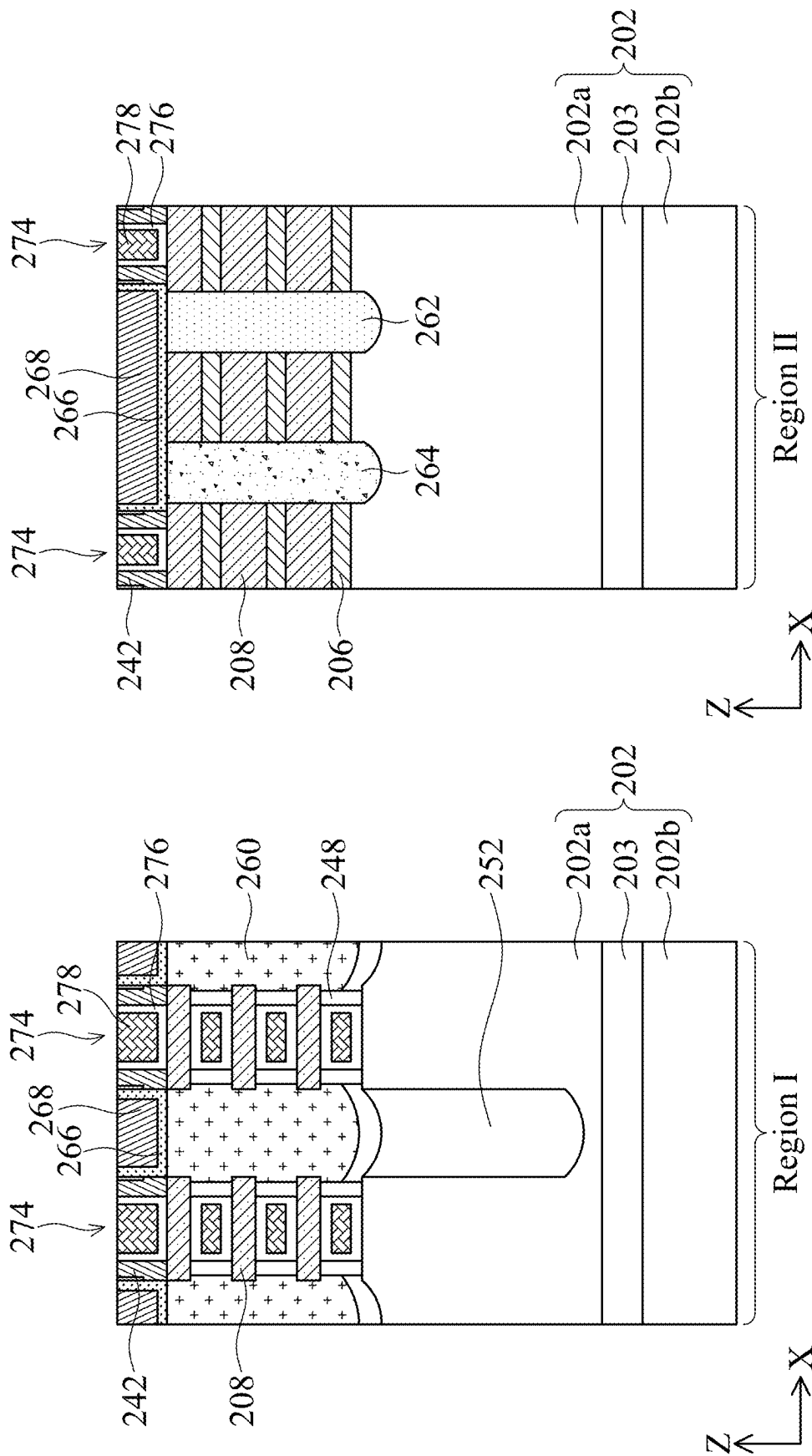


FIG. 14A

FIG. 14B

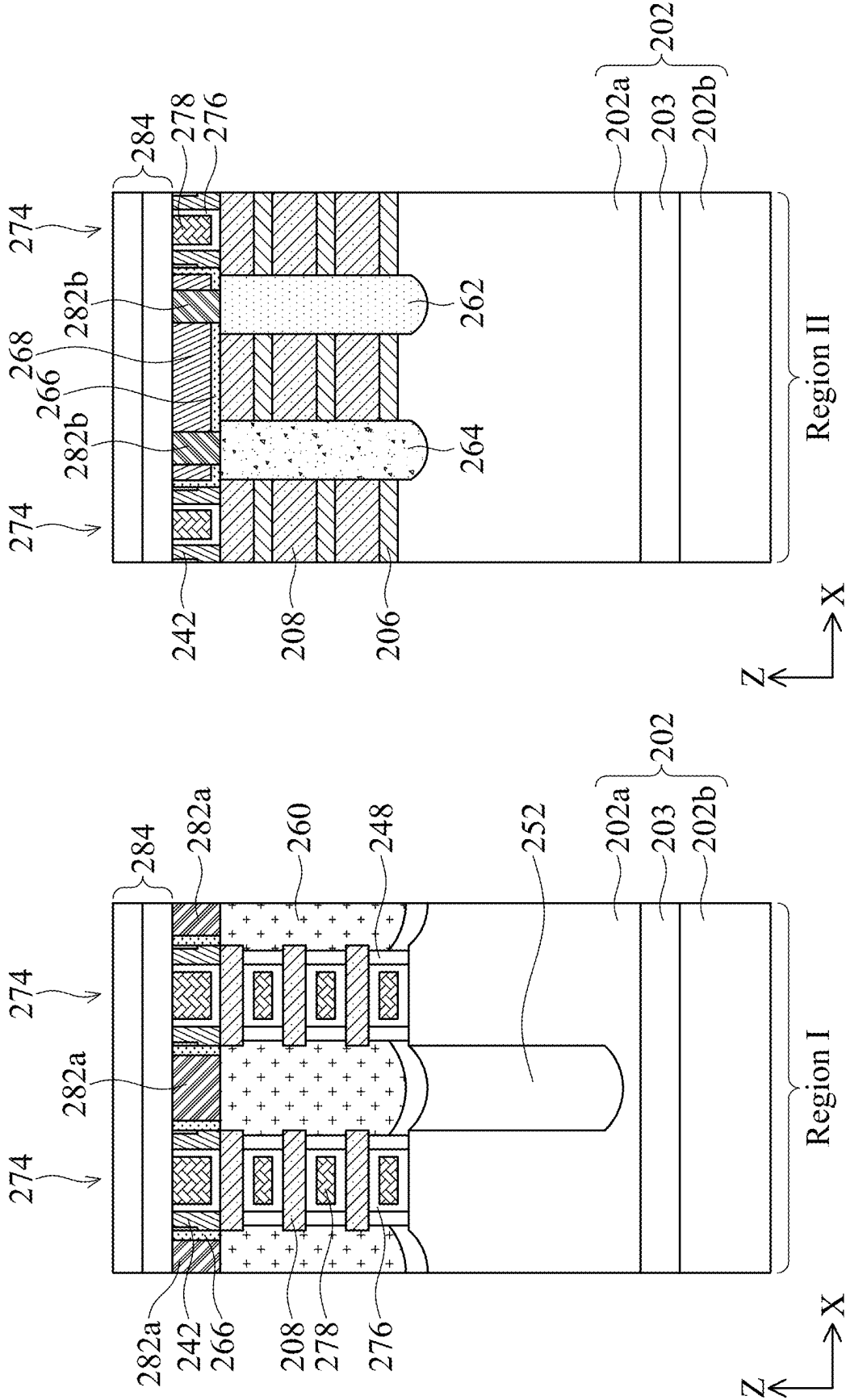


FIG. 15B

FIG. 15A

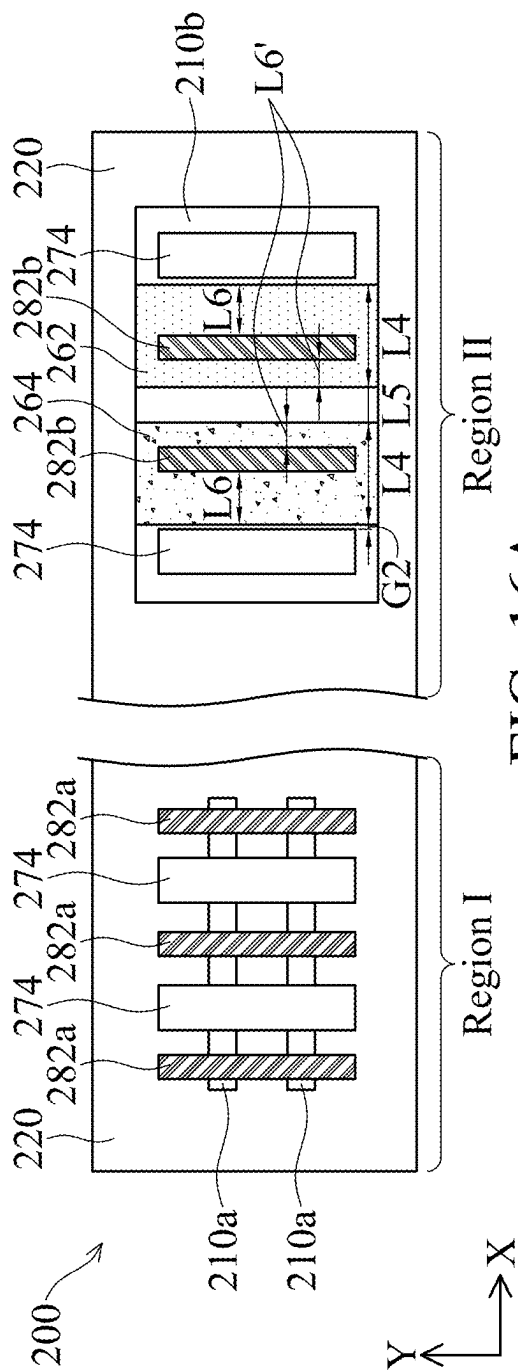


FIG. 16A

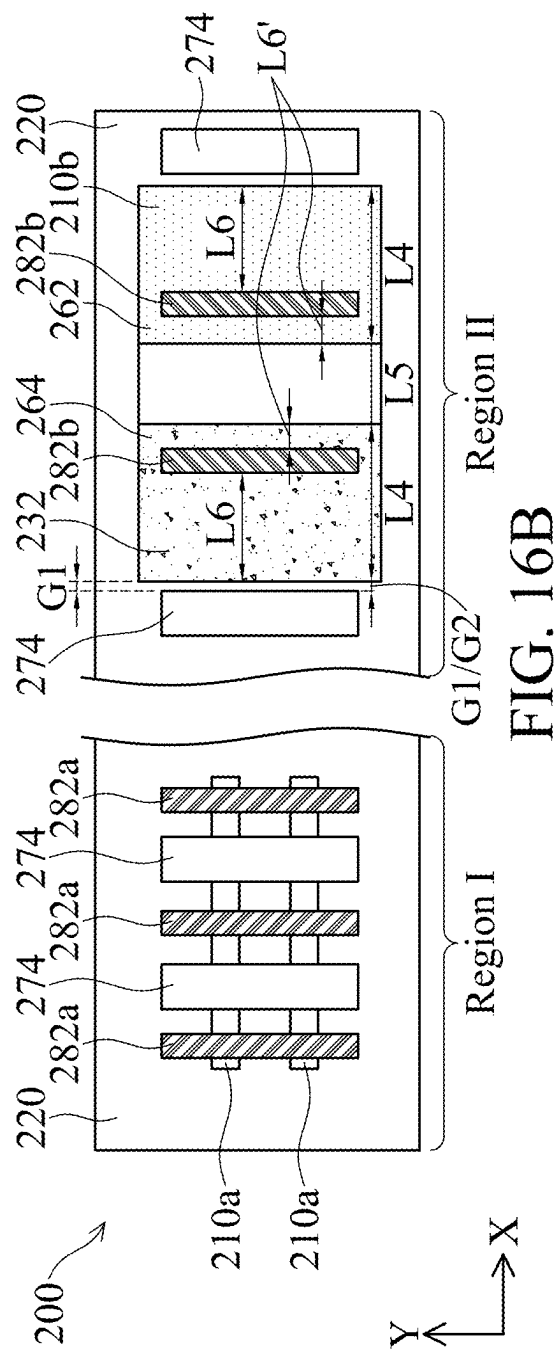


FIG. 16B

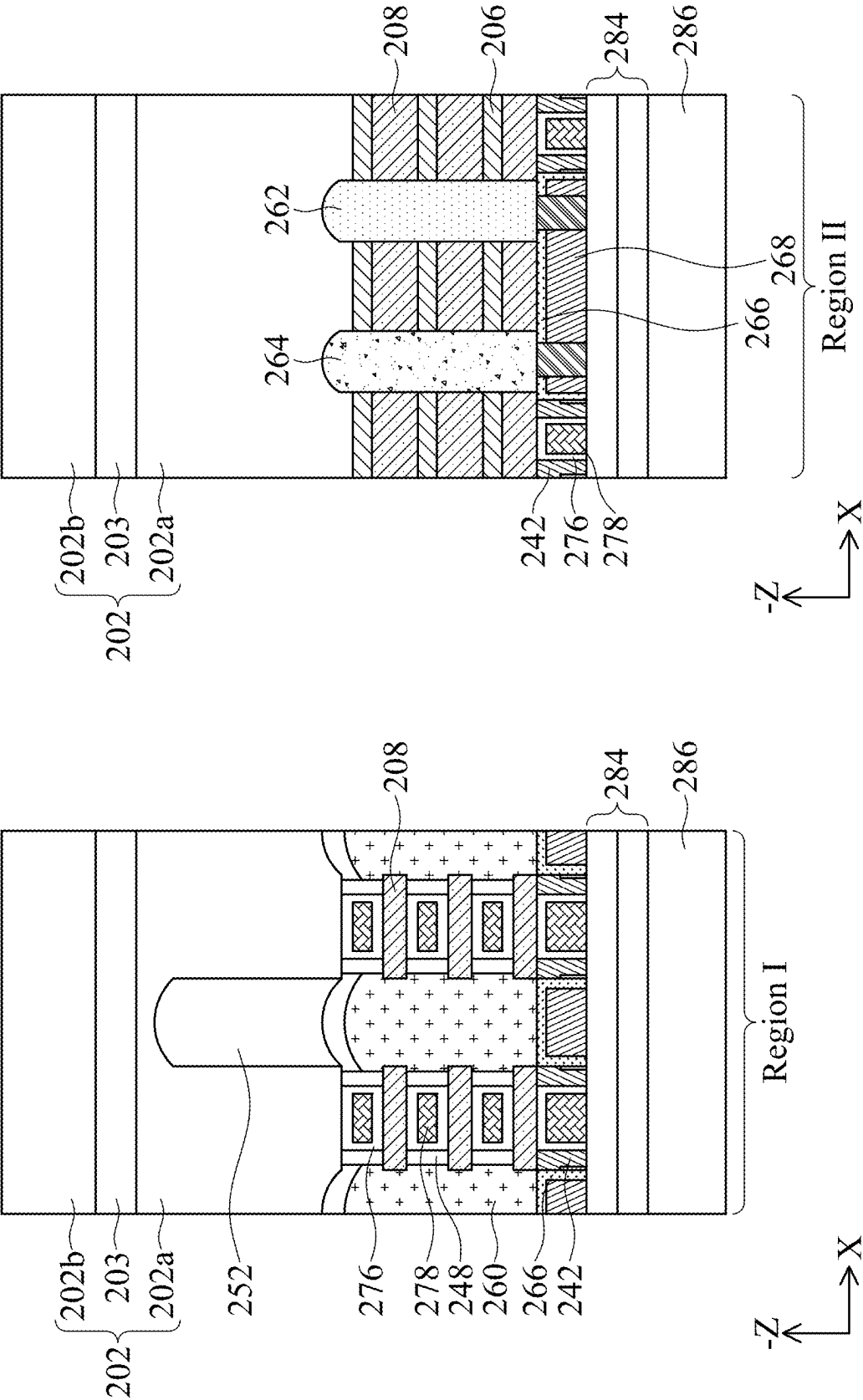


FIG. 17A

FIG. 17B

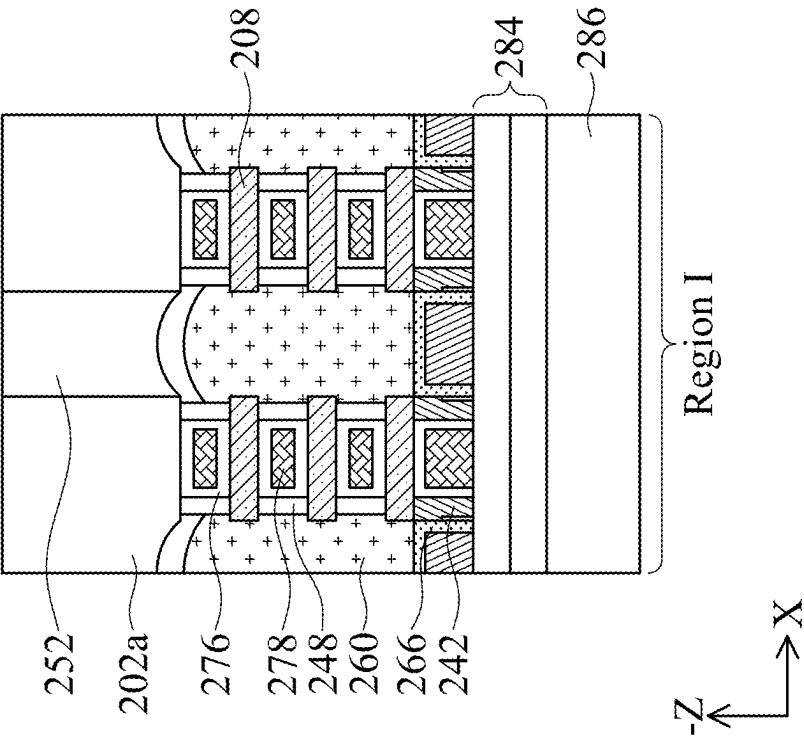


FIG. 18A

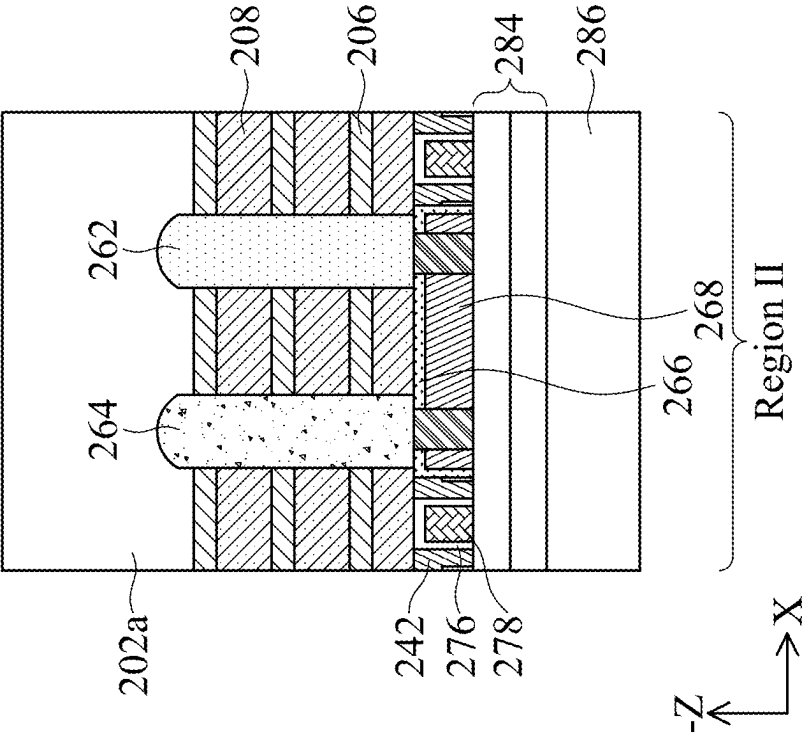


FIG. 18B

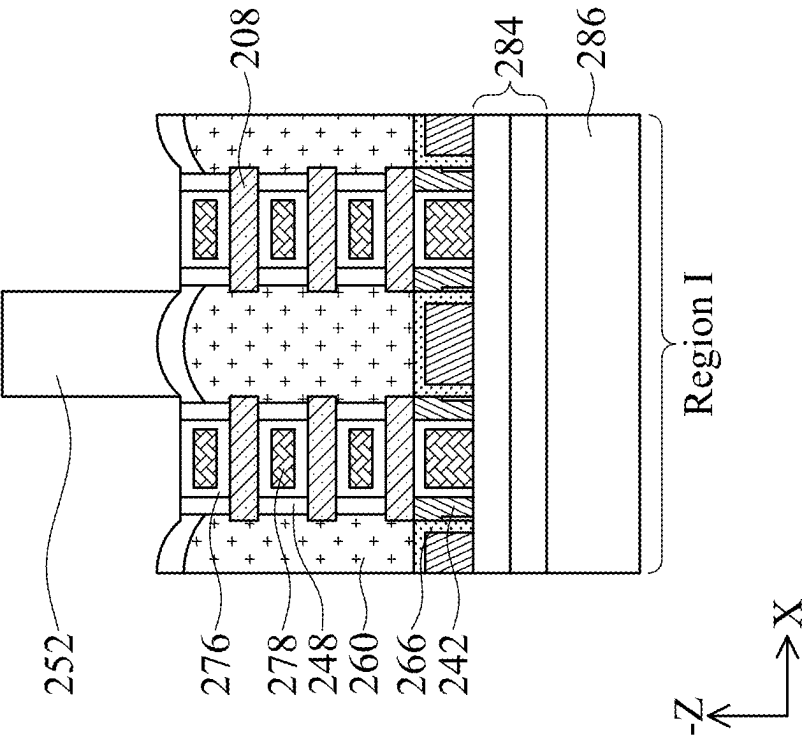


FIG. 19A

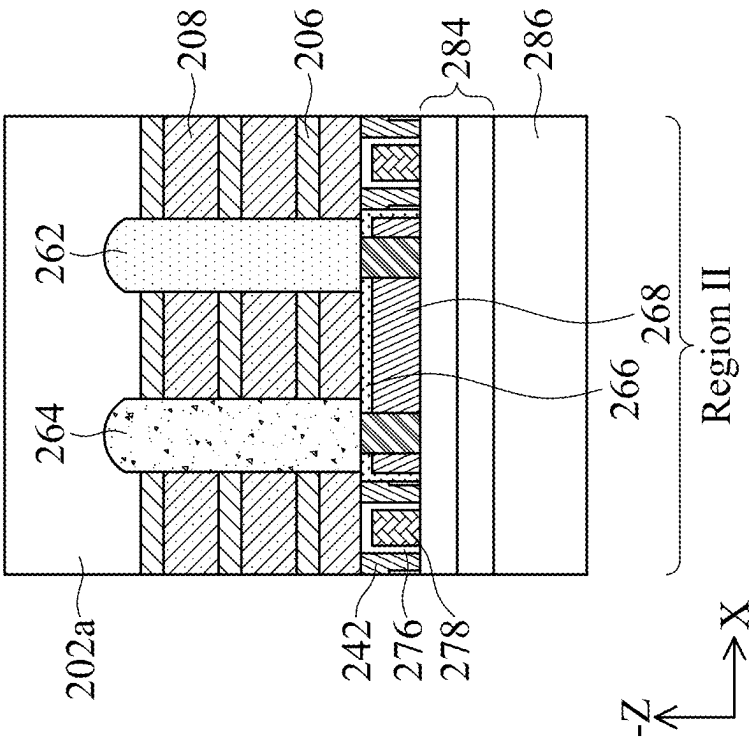


FIG. 19B

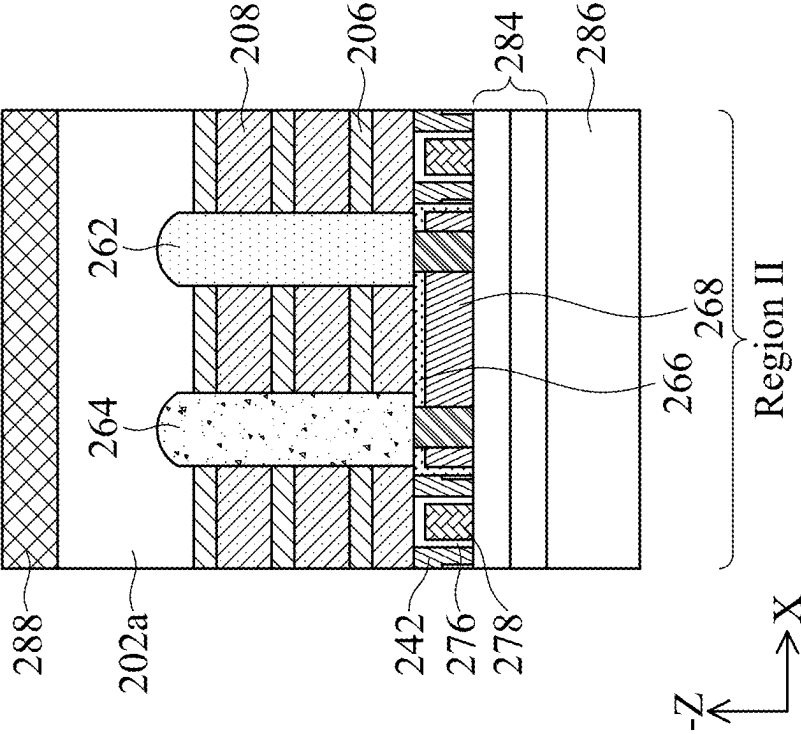


FIG. 20B

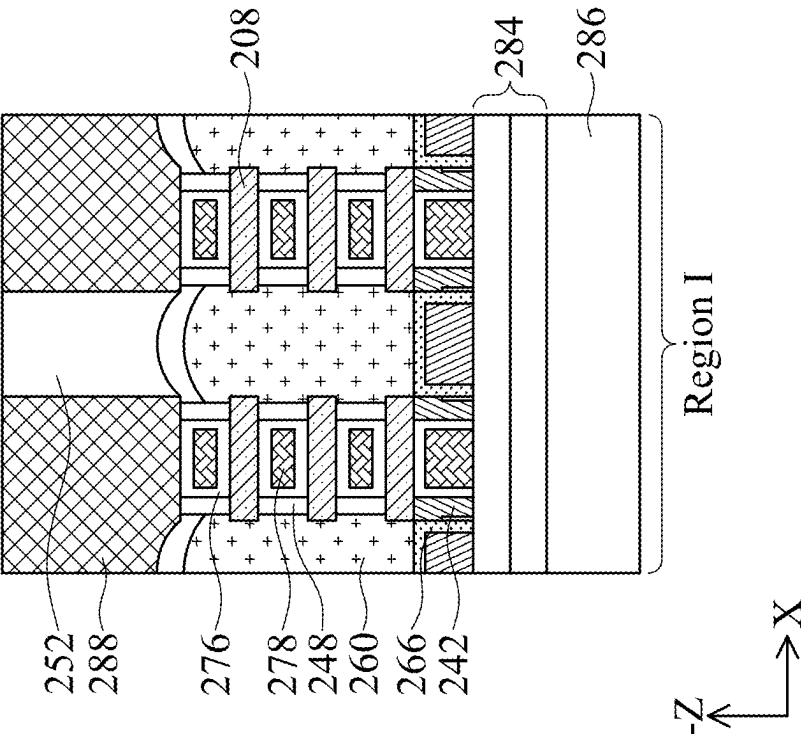


FIG. 20A

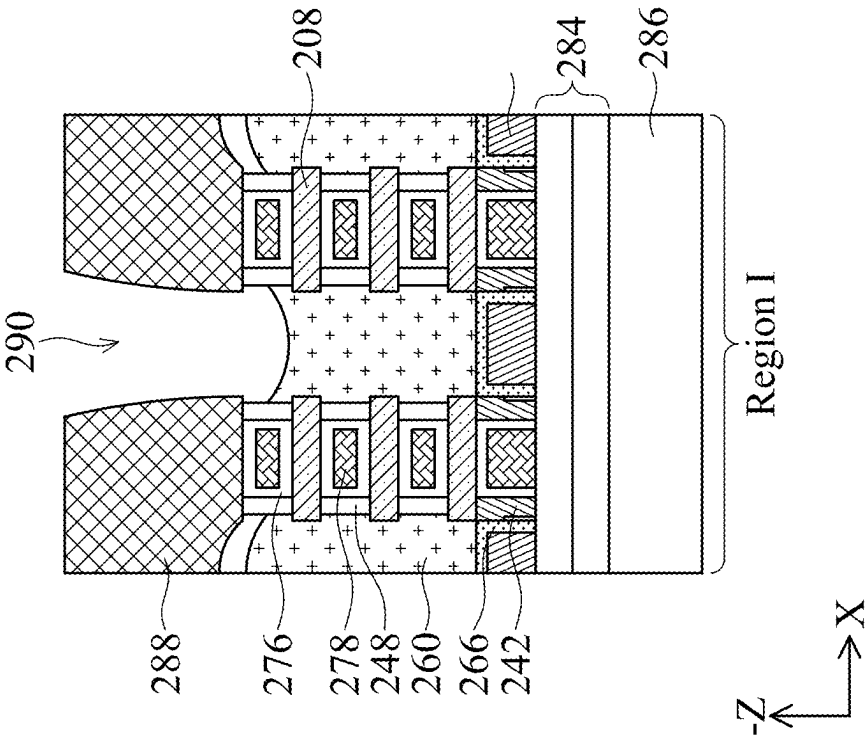


FIG. 21A

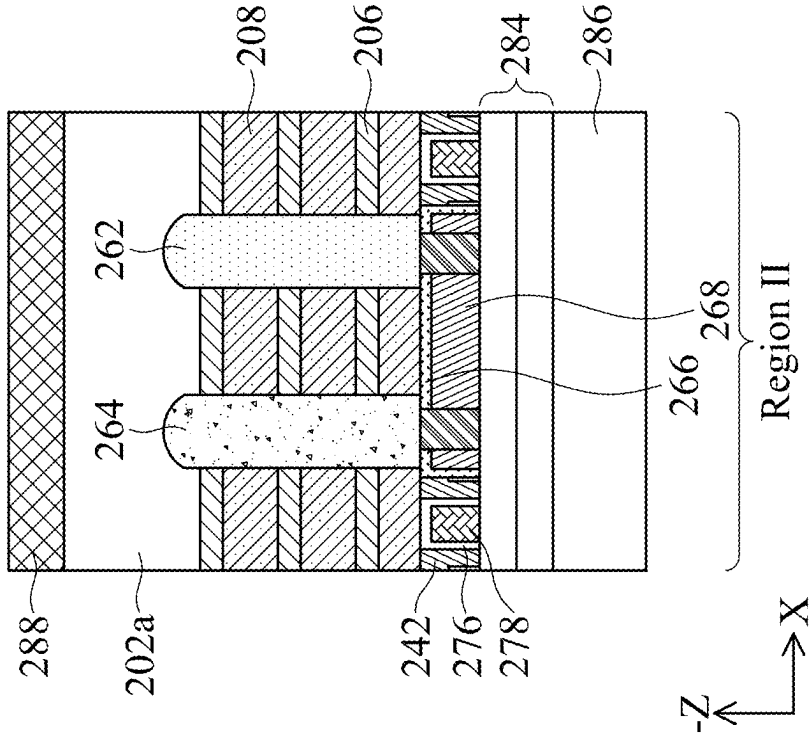


FIG. 21B



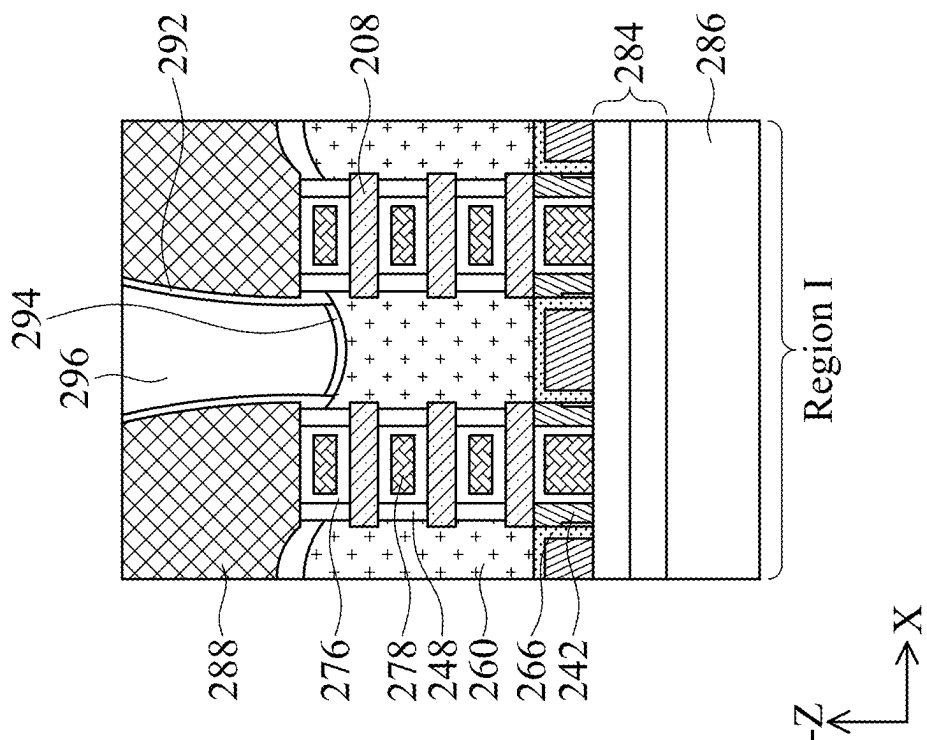


FIG. 22A

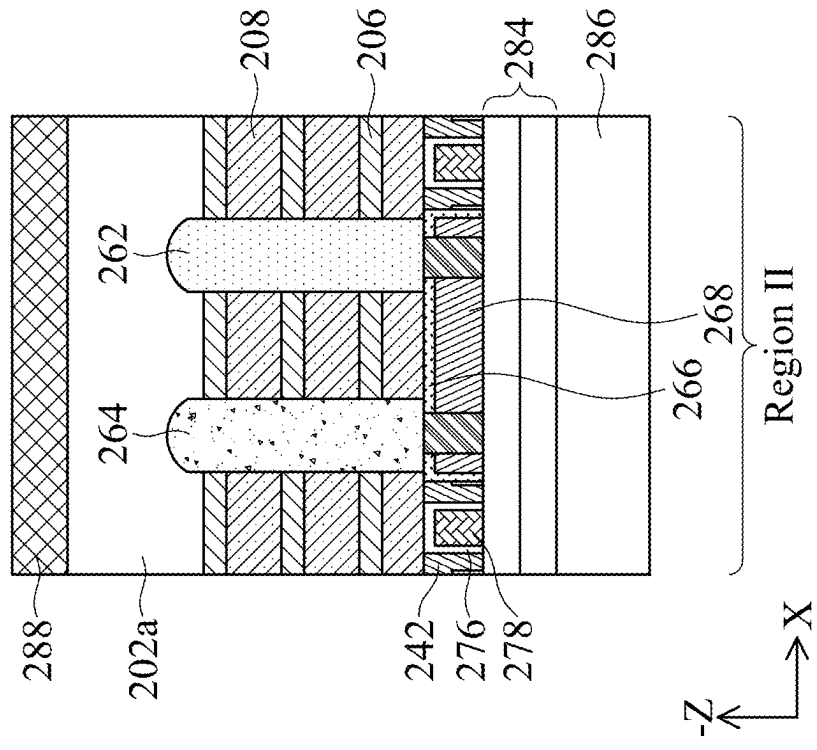
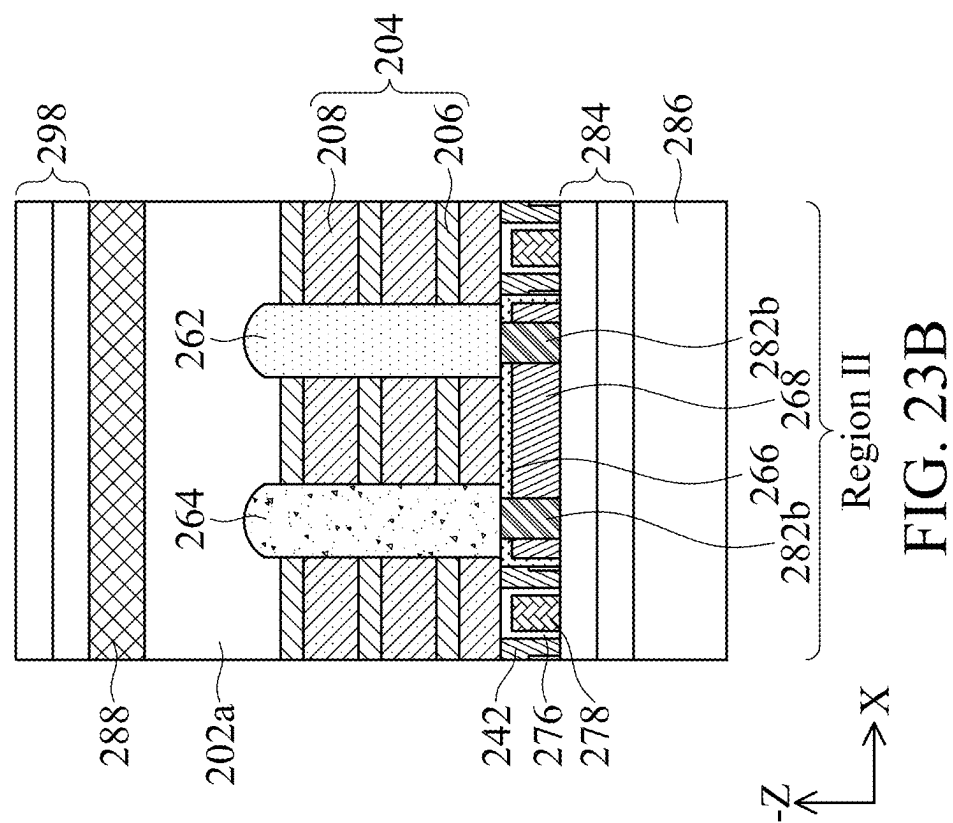
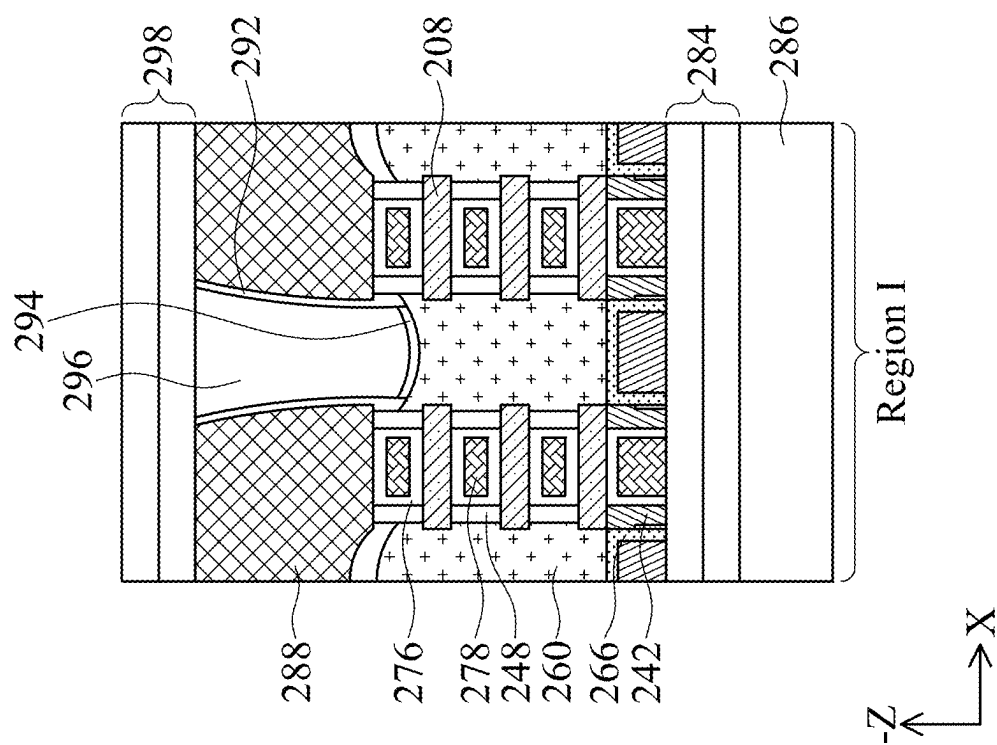


FIG. 22B



## FORMING ESD DEVICES USING MULTI-GATE COMPATIBLE PROCESSES

### PRIORITY

[0001] This is a continuation application of U.S. patent application Ser. No. 18/623,294, filed Apr. 1, 2024, which is a continuation application of U.S. patent application Ser. No. 18/305,556, filed Apr. 24, 2023, now issued as U.S. Pat. No. 11,948,936, which is a continuation application of U.S. patent application Ser. No. 17/224,671, filed Apr. 7, 2021, now issued as U.S. Pat. No. 11,637,099, which claims the benefits to U.S. Provisional Application Ser. No. 63/039,293, filed Jun. 15, 2020, each of which is incorporated herein by reference in its entirety.

### BACKGROUND

[0002] The semiconductor integrated circuit (IC) industry has experienced exponential growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs. Such scaling down has also increased the complexity of processing and manufacturing ICs.

[0003] For example, as IC technologies progress towards smaller technology nodes, multi-gate devices have been introduced to improve gate control by increasing gate-channel coupling, reducing off-state current, and reducing short-channel effects (SCEs). A multi-gate device generally refers to a device having a gate structure, or portion thereof, disposed over more than one side of a channel region. Gate-all-around (GAA) transistors are examples of multi-gate devices that have become popular and promising candidates for high-performance and low-leakage applications. GAA transistors get their name from the gate structure which can extend around the channel region providing access to the stacked semiconductor channel layers on four sides. Compared to planar transistors, such configuration provides better control of the channel and drastically reduces SCEs (in particular, by reducing sub-threshold leakage).

[0004] An IC includes semiconductor devices that serve different functions, such as core functions and electrostatic discharging (ESD) functions. These different functions require semiconductor devices to have different constructions. At the same time, it is advantageous to have similar processes and similar process windows to fabricate these different semiconductor devices to reduce cost and improve yield. Although existing multi-gate (e.g., GAA) transistors and processes are generally adequate for their intended purposes, they have not been entirely satisfactory in every aspect. For example, in the course of IC evolution, backside power rails in addition to frontside interconnects have been introduced to increase the number of metal tracks available for greater power driving capability and simplified metal routing. How to form ESD devices (e.g., ESD diodes) compatible with multi-gate processes that also support backside power rail formation is a challenge faced by the

semiconductor industry. The present disclosure aims to solve the above issues and other related issues.

### BRIEF DESCRIPTION OF THE DRAWINGS

[0005] The present disclosure is best understood from the following detailed description when read with the accompanying figures. It is emphasized that, in accordance with the standard practice in the industry, various features are not drawn to scale and are used for illustration purposes only. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0006] FIGS. 1A, 1B, and 1C show a flow chart of a method of forming a semiconductor structure with ESD devices and backside power rails, according to various aspects of the present disclosure.

[0007] FIGS. 2, 3, and 4 illustrate perspective views of a portion of the semiconductor structure during fabrication processes according to the method of FIGS. 1A-1C, in accordance with some embodiments of the present disclosure.

[0008] FIGS. 5A, 5B, 16A, and 16B illustrate top views of a portion of the semiconductor device during a fabrication process according to the method of FIGS. 1A-1C, in accordance with some embodiments of the present disclosure.

[0009] FIGS. 6A, 7A, 8A, 9A, 10A, 11A, 12A, 13A, 14A, 15A, 17A, 18A, 19A, 20A, 21A, 22A, and 23A illustrate cross-sectional views along the A-A line of a portion of the semiconductor device in FIG. 5A during fabrication processes according to the method of FIGS. 1A-1C, in accordance with some embodiments of the present disclosure.

[0010] FIGS. 6B, 7B, 8B, 9B, 10B, 11B, 12B, 13B, 14B, 15B, 17B, 18B, 19B, 20B, 21B, 22B, and 23B illustrate cross-sectional views along the B-B line of a portion of the semiconductor device in FIG. 5A during fabrication processes according to the method of FIGS. 1A-1C, in accordance with some embodiments of the present disclosure.

### DETAILED DESCRIPTION

[0011] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0012] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90

degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. Still further, when a number or a range of numbers is described with “about,” “approximate,” and the like, the term encompasses numbers that are within certain variations (such as  $\pm 10\%$  or other variations) of the number described, in accordance with the knowledge of the skilled in the art in view of the specific technology disclosed herein, unless otherwise specified. For example, the term “about 5 nm” may encompass the dimensional range from 4.5 nm to 5.5 nm, 4.0 nm to 5.0 nm, etc.

**[0013]** This application generally relates to semiconductor structures and fabrication processes, and more particularly to an electrostatic discharge (ESD) protection device and methods of fabrication thereof that is compatible with multi-gate transistor process flow.

**[0014]** It is well known that extremely high voltages can develop in the vicinity of an integrated circuit (IC) due to the build-up of static charges. A high potential may be generated to an input or output buffer of the integrated circuit, which may be caused by a person touching a package pin that is in electrical contact with the input or output buffer. When the electrostatic charges are discharged, a high current is produced at the package nodes of the integrated circuit, and is referred to electrostatic discharge (ESD). ESD is a serious problem for semiconductor devices since it has the potential of destroying the entire integrated circuit. The duration of the ESD transient is very short, typically in the order of nanoseconds, and the conventional circuit breakers cannot react quickly enough to provide adequate protection. For this reason, it has become a known practice to incorporate ESD devices in ICs. In various embodiments, an IC may have multi-gate transistors (e.g., gate-all-around (GAA) transistors and/or FinFET transistors) placed in a core region providing core functions (e.g., logic and memory applications) and ESD devices placed in an ESD region providing ESD protection.

**[0015]** Yet a recent trend in IC evolution includes providing power rails (or power routings) on a back side (or backside) of a structure containing transistors in addition to an interconnect structure (which may include power rails as well) on a front side (or frontside) of the structure. This increases the number of metal tracks available in the structure for directly connecting to source/drain contacts and vias. It also increases the gate density for greater device integration than existing structures without the backside power rails. The backside power rails may have wider dimension than the first level metal (MO) tracks on the frontside of the structure, which beneficially reduces the power rail resistance. This approach, unfortunately, faces design and process issues. Particularly, a backside thinning process in removing bulk semiconductor substrate substantially eliminates current path for ESD devices comprising multiple fin elements. Accordingly, what is needed in the art is an ESD protection device whose formation process is compatible with the formation of multi-gate process flow while at the same time overcoming the deficiencies of the prior art.

**[0016]** The details of the structure and fabrication methods of the present disclosure are described below in conjunction with the accompanied drawings, which illustrate a process of making a GAA device in a core region and an ESD device in an ESD region, according to some embodiments. A GAA device refers to a device having vertically-stacked horizon-

tally-oriented multi-channel transistors, such as nanowire transistors and nanosheet transistors. GAA devices are promising candidates to take CMOS to the next stage of the roadmap due to their better gate control ability, lower leakage current, and fully FinFET device layout compatibility. For the purposes of simplicity, the present disclosure uses GAA devices as an example. Those of ordinary skill in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures (such as FinFET devices) for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein.

**[0017]** Illustrated in FIGS. 1A-1C is a method **100** of semiconductor fabrication including forming multi-gate transistors together with ESD devices (e.g., ESD diodes). The method **100** is merely an example, and is not intended to limit the present disclosure beyond what is explicitly recited in the claims. Additional operations can be provided before, during, and after the method **100**, and some operations described can be replaced, eliminated, or moved around for additional embodiments of the method. The method **100** is described below in conjunction with FIGS. 2-23B. FIGS. 2-23B have been simplified for the sake of clarity to better understand the inventive concepts of the present disclosure. FIGS. 2-4 illustrate perspective views of a semiconductor device (or device, or structure) **200** according to various stages of the method **100**. FIGS. 5A, 5B, 16A, and 16B illustrate top views of the device **200** according to various stages of the method **100**. FIGS. 6A, 7A, 8A, 9A, 10A, 11A, 12A, 13A, 14A, 15A, 17A, 18A, 19A, 20A, 21A, 22A, and 23A illustrate cross-sectional views along the A-A line of a portion of the device **200** in FIG. 5A which is in a channel region of a GAA transistor and along a lengthwise direction of the channel layers of the GAA transistor, according to various stages of the method **100**. FIGS. 6B, 7B, 8B, 9B, 10B, 11B, 12B, 13B, 14B, 15B, 17B, 18B, 19B, 20B, 21B, 22B, and 23B illustrate cross-sectional views along the B-B line of a portion of the device **200** in FIG. 5A which is along a lengthwise direction of a fin element (or fin) in an ESD device.

**[0018]** In some embodiments, the device **200** is a portion of an IC chip, a system on chip (SoC), or portion thereof, that includes various passive and active microelectronic devices such as resistors, capacitors, inductors, diodes, p-type field effect transistors (PFETs), n-type field effect transistors (NFETs), FinFET, nanosheet FETs, nanowire FETs, other types of multi-gate FETs, metal-oxide semiconductor field effect transistors (MOSFETs), complementary metal-oxide semiconductor (CMOS) transistors, bipolar junction transistors (BJTs), laterally diffused MOS (LD-MOS) transistors, high voltage transistors, high frequency transistors, memory devices, other suitable components, or combinations thereof. Additional features can be added in the device **200**, and some of the features described below can be replaced, modified, or eliminated in other embodiments of the device **200**.

**[0019]** The method **100** at operation **102** (FIG. 1A) provides (or is provided with) the device **200**. Referring to FIG. 2, the device **200** includes a region I for forming transistors and a region II for forming ESD devices. The transistors formed in the region I are for core applications such as for forming logic circuits, memory circuits, and other core circuits. The ESD devices, such as ESD diodes, formed in the region II are for protecting the device **200** from electro-

static discharging events. The device **200** includes a substrate **202** and an epitaxial stack **204** above the substrate **202**. Both the substrate **202** and the epitaxial stack **204** extend continuously from the region I to the region II. Further, a hard mask (HM) layer **212** is formed over the epitaxial stack **204**.

**[0020]** In some embodiments, the substrate **202** is a semiconductor substrate such as a silicon substrate. The substrate **202** may include various layers, including conductive or insulating layers formed on a semiconductor substrate. In the illustrated embodiment, the substrate **202** includes a dielectric capping layer **203**. In some embodiments, the dielectric capping layer **203** is an oxide layer. The dielectric capping layer **203** may be deposited using chemical vapor deposition (CVD), atomic layer deposition (ALD), physical vapor deposition (PVD), or oxidation. The dielectric capping layer **203** separates the substrate **202** into a top substrate portion **202a** and a bottom substrate portion **202b**. In some embodiments, the dielectric capping layer **203** serves as an etch stop or a chemical mechanical polishing (CMP) stop during a backside thinning process.

**[0021]** The substrate **202** may also include various doping configurations depending on design requirements as is known in the art. For example, different doping profiles (e.g., n-wells, p-wells) may be formed on the substrate **202** in regions designed for different device types (e.g., n-type field effect transistors (N-FET), p-type field effect transistors (P-FET)). The suitable doping may include ion implantation of dopants and/or diffusion processes. The substrate **202** may have isolation features (e.g., shallow trench isolation (STI) features) interposing the regions providing different device types. The substrate **202** may also include other semiconductors such as germanium, silicon carbide (SiC), silicon germanium (SiGe), or diamond. Alternatively, the substrate **202** may include a compound semiconductor and/or an alloy semiconductor. Further, the substrate **202** may optionally include an epitaxial layer (epi-layer), may be strained for performance enhancement, may include a silicon-on-insulator (SOI) structure, and/or may have other suitable enhancement features.

**[0022]** The epitaxial stack **204** includes epitaxial layers **206** of a first composition interposed by epitaxial layers **208** of a second composition. The first and second compositions can be different. The epitaxial layers **208** may include the same composition as the substrate **202**. In the illustrated embodiment, the epitaxial layers **206** are silicon germanium (SiGe) and the epitaxial layers **208** are silicon (Si). However, other embodiments are possible including those that provide for a first composition and a second composition having different oxidation rates and/or etch selectivity. For example, in some embodiments, either of the epitaxial layers **206**, **208** of the first composition or the second composition may include other materials such as germanium, a compound semiconductor such as silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide, an alloy semiconductor such as SiGe, GaAsP, AlInAs, AlGaAs, InGaAs, GaInP, and/or GaInAsP, or combinations thereof. In some embodiments, the epitaxial layers **206** and **208** are substantially dopant-free (i.e., having an extrinsic dopant concentration from about  $0\text{ cm}^{-3}$  to about  $1 \times 10^{17}\text{ cm}^{-3}$ ), where for example, no intentional doping is performed during the epitaxial growth process. By way of example, epitaxial growth of the epitaxial layers **206**, **208** of the first composition or the second

composition may be performed by a molecular beam epitaxy (MBE) process, a metalorganic chemical vapor deposition (MOCVD) process, and/or other suitable epitaxial growth processes. In various embodiments, the substrate **202** is a crystalline substrate, the epitaxial layers **206**, **208** are crystalline semiconductor layers, and the epitaxial stack **204** is formed as a superlattice structure.

**[0023]** In some embodiments, each epitaxial layer **206** has a thickness ranging from about 2 nanometers (nm) to about 6 nm. The epitaxial layers **206** may be substantially uniform in thickness. Yet in the illustrated embodiment, the top epitaxial layer **206** is thinner (e.g., half the thickness) than other epitaxial layers **206** thereunder. In some embodiments, each epitaxial layer **208** has a thickness ranging from about 6 nm to about 12 nm. In some embodiments, the epitaxial layers **208** of the stack are substantially uniform in thickness. As described in more detail below, the epitaxial layers **208** or portions thereof may form channel layer(s) of the subsequently-formed GAA transistors in the region I and the thickness is chosen based on transistor performance considerations. The term channel layer(s) is used herein to designate any material portion for channel(s) in a transistor with nanoscale, or even microscale dimensions, and having an elongate shape, regardless of the cross-sectional shape of this portion. Thus, this term designates both circular and substantially circular cross-section elongate material portions, and beam or bar-shaped material portions including for example a cylindrical in shape or substantially rectangular cross-section. The epitaxial layers **206** in channel regions(s) may eventually be removed and serve to define a vertical distance between adjacent channel members for a subsequently-formed multi-gate device and the thickness is chosen based on device performance considerations. Accordingly, the epitaxial layers **206** may also be referred to as sacrificial layers, and epitaxial layers **208** may also be referred to as channel layers.

**[0024]** It is noted that three (3) layers of the epitaxial layers **206** and three (3) layers of the epitaxial layers **208** are alternately arranged as illustrated in FIG. 2, which is for illustrative purposes only and not intended to be limiting beyond what is specifically recited in the claims. It can be appreciated that any number of epitaxial layers can be formed in the epitaxial stack **204**; the number of layers depending on the desired number of channels layers for the device **200**. In some embodiments, the number of epitaxial layers **208** is between 2 and 10. It is also noted that while the epitaxial layers **206**, **208** are shown as having a particular stacking sequence, where an epitaxial layer **208** is the topmost layer of the epitaxial stack **204**, other configurations are possible. For example, in some cases, an epitaxial layer **206** may alternatively be the topmost layer of the epitaxial stack **204**. Stated another way, the order of growth for the epitaxial layers **206**, **208**, and thus their stacking sequence, may be switched or otherwise be different than what is shown in the figures, while remaining within the scope of the present disclosure.

**[0025]** In the illustrated embodiment, the hard mask (HM) layer **212** is formed over the epitaxial stack **204** prior to patterning in forming semiconductor fins. In some embodiments, the HM layer **212** includes an oxide layer **212A** (e.g., a pad oxide layer that may include  $\text{SiO}_2$ ) and a nitride layer **212B** (e.g., a pad nitride layer that may include  $\text{Si}_3\text{N}_4$ ) formed over the oxide layer **212A**. The oxide layer **212A** may act as an adhesion layer between the epitaxial stack **204**

and the nitride layer **212B** and may act as an etch stop layer for etching the nitride layer **212B**. In some examples, the HM layer **212** includes thermally grown oxide, CVD-deposited oxide, and/or ALD-deposited oxide. In some embodiments, the HM layer **212** includes a nitride layer deposited by CVD and/or other suitable technique.

[0026] The method **100** then proceeds to operation **104** (FIG. 1A) where fin elements (or fins) **210** are formed by patterning the epitaxial stack **204** by using the patterned mask layer **212**. With reference to the example of FIG. 3, in an embodiment of operation **104**, a plurality of fins **210** extending from the substrate **202** are formed extending lengthwise in the X direction, including fins **210a** in the region I and fins **210b** in the region II. In the illustrated embodiment, two fins **210a** are arranged in the Y direction in the region I and one fin **210b** is depicted in the region II. But the number of the fins **210a** or **210b** is not limited to and may be as small as one or three or more. In various embodiments, each of the fins **210** includes an upper portion constituted by the stacked epitaxial layers **206** and **208**, and a lower portion formed from the substrate **202**. The width **W1** of the upper portion of the fins **210a** in the region I along the Y direction is in a range from about 10 nm to about 40 nm in some embodiments. The width **W2** of the upper portion of the fins **210b** in the region II along the Y direction is wider than **W1**, such as in a range from about 50 nm to about 1000 nm in some embodiments. The length **L1** of the upper portion of the fins **210a** in the region I along the X direction and the length **L2** of the upper portion of the fins **210b** in the region II along the X direction may each range from about 90 nm to about 900 nm in some embodiments. In the region I, the fins **210a** may be designed for forming an NFET or a PFET. In the region II, the fins **210b** may be designed for forming an ESD diode. A large width **W2** of the fins **210b** provides lower ESD resistance and stronger ESD current conductivity.

[0027] The fins **210** may be fabricated using suitable processes including photolithography and etch processes. The photolithography process may include forming a photoresist layer (not shown) over the HM layer **212**, exposing the resist to a pattern, performing post-exposure bake processes, and developing the resist to form a masking element including the resist. In some embodiments, patterning the resist to form the masking element may be performed using an electron beam (e-beam) lithography process. The masking element may then be used to protect regions of the substrate **202**, and layers formed thereupon, while an etch process forms trenches **214** in unprotected regions through the HM layer **212**, through the epitaxial stack **204**, and into the substrate **202**, thereby leaving the plurality of extending fins **210**. The trenches **214** may be etched using dry etching, wet etching, RIE, and/or other suitable processes.

[0028] Numerous other embodiments of methods to form the semiconductor fins on the substrate may also be used including, for example, defining the fin region (e.g., by mask or isolation regions) and epitaxially growing the epitaxial stack **204** in the form of the fins **210**. In some embodiments, forming the fins **210** may include a trim process to decrease the width of the fins **210**. The trim process may include wet and/or dry etching processes. In some embodiments, the fins **210** may be fabricated using suitable processes including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing pat-

terns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers, or mandrels, may then be used to pattern the fins **210** by etching the epitaxial stack **204** and the substrate **202**. The etching process can include dry etching, wet etching, reactive ion etching (RIE), and/or other suitable processes.

[0029] At operation **106**, the method **100** (FIG. 1A) forms isolation structures, such as shallow trench isolation (STI) features, between the fins **210**. Still referring to FIG. 3, STI features **220** is disposed on the substrate **202** interposing the fins **210**. By way of example, in some embodiments, a dielectric layer is first deposited over the substrate **202**, filling the trenches **214** with dielectric material. In some embodiments, the dielectric layer may include silicon oxide, silicon nitride, silicon oxynitride, fluorine-doped silicate glass (FSG), a low-k dielectric, combinations thereof, and/or other suitable materials. In various examples, the dielectric layer may be deposited by a CVD process, a SACVD process, a flowable CVD process, an ALD process, a PVD process, and/or other suitable process. In some embodiments, after deposition of the dielectric layer, the device **200** may be annealed, for example, to improve the quality of the dielectric layer. In some embodiments, the dielectric layer may include a multi-layer structure, for example, having one or more liner layers. In some embodiments of forming the STI features **220**, after deposition of the dielectric layer, the deposited dielectric material is thinned and planarized, for example by a chemical mechanical polishing (CMP) process, such that the upper surface of the topmost epitaxial layer **208** is exposed. Subsequently, the dielectric layer interposing the fins **210** are recessed. As shown in FIG. 3, the STI features **220** are recessed providing the fins **210** extending above the STI features **220**. In some embodiments, the recessing process may include a dry etching process, a wet etching process, and/or a combination thereof. In some embodiments, a recessing depth is controlled (e.g., by controlling an etching time) so as to result in a desired height of the exposed upper portion of the fins **210**. In the illustrated embodiment, each of the layers of the epitaxial stack **204** is exposed. In furtherance of the embodiment, a top surface of the STI features **220** is recessed below a bottom surface of the epitaxial stack **204**.

[0030] The method **100** then proceeds to operation **108** (FIG. 1A) where sacrificial layers/features are formed and in particular, dummy gate structures (or dummy gate stacks). While the present discussion is directed to a replacement gate (or gate-last) process whereby a dummy gate structure is formed and subsequently replaced, other configurations may be possible.

[0031] Referring to FIGS. 4, 5A-B, and 6A-B, gate stacks **232** are formed. FIG. 4 is a perspective view of the device **200**, FIGS. 5A and 5B refer to top views of two embodiments of the device **200**, FIG. 6A refers to a cross-sectional view in the region I taken in a fin **210a** and along the lengthwise direction of the fin **210a** (e.g., along the A-A line in FIG. 4), and FIG. 6B refers to a cross-sectional view in the region II taken in the fin **210b** and along the lengthwise direction of the fin **210b** (e.g., along the B-B line in FIG. 4). The gate stacks **232** are oriented lengthwise along the Y

direction. The gate stacks **232** have a width **W3** in a range from about 10 nm to about 100 nm and a length **L3** in a range from about 40 nm to about 950 nm, in some embodiments. In the region I, the length **L3** of the gate stacks **232** is larger than the width **W1** of the fins **210a**, and the gate stacks **232** engages the fins **210** from both the top surface and sidewalls of the fins **210a**. The portion of the fins **210a** underlying the dummy gate stacks **232** may be referred to as the channel region. The dummy gate stacks **232** may also define source/drain (S/D) regions of the fins **210a**, for example, the regions of the fins **210a** adjacent and on opposing sides of the channel region. A distance **D1** between adjacent gate stacks **232** may range from about 30 nm to about 200 nm. In the region II, the length **L3** of the gate stacks **232** is smaller than the width **W2** of the fin **210b**, and the gate stacks **232** are deposited above the top surface of the fin **210b** but not on its sidewalls (FIG. 5A). A distance **D2** between two adjacent gate stacks **232** in the region II is in a range from about 110 nm to about 1020 nm, which is larger than **D1**, in some embodiments. In some embodiments, a ratio of **D2/L2** is not less than about 0.7. If the ratio of **D2/L2** is less than 70%, the relative close distance between two adjacent gate stacks **232** may introduce unwanted stray capacitance and degrade ESD performance. In furtherance of some embodiments, the ratio of **D2/L2** is larger than 1, which is that the distance **D2** being larger than the length **L2** of the fin **210b** and the gate stacks **232** being deposited on the STI features **220** that surrounds the fin **210b** (FIG. 5B). A gap **G1** between two opposing edges of the gate stacks **232** and the fin **210b** may range from about 10 nm to about 60 nm.

[0032] In various embodiments, the gate stacks **232** are a dummy (sacrificial) gate stacks that are subsequently removed. Thus, in some embodiments using a gate-last process, the gate stacks **232** are dummy gate stacks and will be replaced by the final gate stack at a subsequent processing stage of the device **200**. In particular, the dummy gate stacks **232** may be replaced at a later processing stage by a high-K dielectric layer (HK) and metal gate electrode (MG) as will be discussed in more detail below.

[0033] In some embodiments, the dummy gate stack **232** includes a dummy dielectric layer **234** and a dummy electrode layer **236**. In some embodiments, the dummy dielectric layer **234** may include SiO<sub>2</sub>, silicon nitride, a high-K dielectric material and/or other suitable material. In various examples, the dummy dielectric layer **234** may be deposited by a CVD process, a subatmospheric CVD (SACVD) process, a flowable CVD process, an ALD process, a PVD process, or other suitable process. By way of example, the dummy dielectric layer **234** may be used to prevent damages to the fins **210** by subsequent processes (e.g., subsequent formation of the dummy gate stack). Subsequently, other portions of the dummy gate stack **232** are formed, including a dummy electrode layer **236** and a hard mask **240** which may include multiple layers **240A** and **240B** (e.g., an oxide layer **240A** and a nitride layer **240B**). In some embodiments, the dummy gate stacks **232** is formed by various process steps such as layer deposition, patterning, etching, as well as other suitable processing steps. Exemplary layer deposition processes include CVD (including both low-pressure CVD and plasma-enhanced CVD), PVD, ALD, thermal oxidation, e-beam evaporation, or other suitable deposition techniques, or combinations thereof. In forming the gate stack for example, the patterning process includes a lithography process (e.g., photolithography or e-beam lithography) which

may further include photoresist coating (e.g., spin-on coating), soft baking, mask aligning, exposure, post-exposure baking, photoresist developing, rinsing, drying (e.g., spin-drying and/or hard baking), other suitable lithography techniques, and/or combinations thereof. In some embodiments, the etching process may include dry etching (e.g., RIE etching), wet etching, and/or other etching methods. In some embodiments, the dummy electrode layer **236** may include polycrystalline silicon (polysilicon). In some embodiments, the hard mask **240** includes an oxide layer **240A** such as a pad oxide layer that may include SiO<sub>2</sub>. In some embodiments, hard mask **240** includes a nitride layer **240B** such as a pad nitride layer that may include Si<sub>3</sub>N<sub>4</sub>, silicon oxynitride and/or silicon carbide.

[0034] In the following figures, for the sake of conciseness, the manufacturing operations after the structure shown in FIG. 5A is formed, in which the dummy gate stacks are deposited directly above the fin **210b**, are explained. However, the same operations can be applied to the structure shown in FIG. 5B, in which the dummy gate stacks are deposited directly above the STI features **220**.

[0035] At operation **110**, the method **100** (FIG. 1A) forms gate spacers **242** on sidewalls of the dummy gate stacks **232**. Referring to FIGS. 7A and 7B, the gate spacers **242** may have a thickness of about 2-10 nm. In some examples, the gate spacers **242** may include a dielectric material such as silicon oxide, silicon nitride, silicon carbide, silicon oxynitride, SiCN, silicon oxycarbide, SiOCN, a low-K material, and/or combinations thereof. In some embodiments, the gate spacers **242** include multiple layers, such as a liner spacer layer **242A** and a main spacer layer **242B**, and the like. By way of example, the gate spacers **242** may be formed by conformally depositing a dielectric material over the device **200** using processes such as a CVD process, a subatmospheric CVD (SACVD) process, a flowable CVD process, an ALD process, a PVD process, or other suitable process. Following the conformal deposition of the dielectric material, portions of the dielectric material used to form the gate spacers **242** may be etched-back to expose portions of the fins **210** not covered by the dummy gate stacks **232**. In some cases, the etch-back process removes portions of dielectric material used to form the gate spacers **242** along a top surface of the dummy gate stacks **232**. In some embodiments, the etch-back process may include a wet etch process, a dry etch process, a multiple-step etch process, and/or a combination thereof. It is noted that after the etch-back process, the gate spacers **242** remain disposed on sidewalls of the dummy gate stacks **232**.

[0036] At operation **112**, the method **100** (FIG. 1A) etches the S/D regions in the region I to form recesses **246**. The recesses **246** are also referred to as S/D trenches **246**. Referring to FIGS. 8A and 8B, in some embodiments of operation **112**, an etch mask (not shown) is formed above the device **200**. The etch mask provides openings over the region I, allowing the S/D regions of the fins **210a** to be recessed through the openings, while protecting the region II from etching loss. The operation **112** then applies an etching process that is tuned to be selective to the materials of the semiconductor material (e.g. silicon and silicon germanium) in the fins **210a** and with no (or minimal) etching to the dummy gate stacks **232** and the STI features **220**. In the present embodiment, the etching process recesses the fins **210a** in the S/D regions to a level that is below the bottommost epitaxial layer **206**. Terminal ends of the epi-

taxial layers **206** and **208** are exposed in the S/D trenches **246**. The etching process can be dry etching, wet etching, reactive ion etching, or other etching methods. The etch mask is subsequently removed, for example, by a resist stripping process or other suitable process.

**[0037]** At operation **114**, the method **100** (FIG. 1A) forms inner spacers **248** on terminal ends of the epitaxial layer **206** in the region I. Referring to FIGS. 9A and 9B, in some embodiments, the inner spacers **248** include a dielectric material that includes silicon, oxygen, carbon, nitrogen, other suitable material, or combinations thereof (for example, silicon oxide, silicon nitride, silicon oxynitride, silicon carbide, or silicon oxycarbonitride). The inner spacers **248** may be formed by deposition and etching processes. For example, after the S/D trenches **246** are formed, a lateral etch process may be used to recess the epitaxial layers **206** to form cavities vertically between the adjacent epitaxial layers **208**. Then, one or more dielectric materials are deposited (using CVD or ALD for example) to fill the cavities. Another etching process is performed to remove the dielectric materials outside the cavities, thereby forming the inner spacers **248**.

**[0038]** At operation **116**, the method **100** (FIG. 1A) further extends downwardly an S/D trench **246** in a source region (thereby as a source trench **246**) into the top substrate portion **202a** in an etch process and forms a bottom S/D layer **252** in the source trench **246**. The resultant structure is shown in FIGS. 10A and 10B according to an embodiment. In the illustrated embodiment, the operation **114** first forms an etch mask **250** over the device **200**. In some embodiments, the etch mask **250** is a liner layer conformally deposited on the device **200**. The etch mask **250** provides an opening in the region I over a source trench **246** between adjacent dummy gate stacks **232**, while drain trenches **246** and the region II remain covered. In various embodiments, the opening of the etch mask **250** may be provided over the drain trench only, source trench only, or both source and drain trenches. In the present disclosure, a source and a drain are interchangeably used. The etch mask **250** includes a material that is different than a semiconductor material in the top substrate portion **202a** to achieve etching selectivity during further etching the source trench **246**. For example, the etch mask **250** includes a resist material (and thus may be referred to as a patterned resist layer and/or a patterned photoresist layer). The operation **116** further includes etching the top substrate portion **202a** through the etch mask **250** to extend downwardly the source trench **246**. The etching process can be dry etching, wet etching, reactive ion etching, or other suitable etching methods, to selectively recess the source trench **246** to the desired level. The operation **116** subsequently fills the bottom portion of the extended source trench **246** with a bottom S/D layer **252** that has etching selectivity with the semiconductor material in the top substrate portion **202a**. In some embodiments, the bottom S/D layer **252** is a dielectric layer that may include one or more dielectric material selected from  $\text{La}_2\text{O}_3$ ,  $\text{Al}_2\text{O}_3$ ,  $\text{SiOCN}$ ,  $\text{SiOC}$ ,  $\text{SiCN}$ ,  $\text{SiO}_2$ ,  $\text{SiC}$ ,  $\text{ZnO}$ ,  $\text{ZrN}$ ,  $\text{Zr}_2\text{Al}_3\text{O}_9$ ,  $\text{TiO}_2$ ,  $\text{TaO}_2$ ,  $\text{ZrO}_2$ ,  $\text{HfO}_2$ ,  $\text{Si}_3\text{N}_4$ ,  $\text{Y}_2\text{O}_3$ ,  $\text{AlON}$ ,  $\text{TaCN}$ ,  $\text{ZrSi}$ , combinations thereof, and other suitable material(s), and may be formed by PE-CVD, F-CVD or other suitable methods. In some embodiments, the bottom S/D layer **252** may include semiconductor material(s) so long as etching selectivity is achieved between the bottom S/D layer **252** and the top substrate portion **202a**. For example, the bottom S/D layer

**252** may include SiGe, and may be formed by any epitaxy processes including chemical vapor deposition (CVD) techniques (for example, vapor phase epitaxy and/or Ultra-High Vacuum CVD), molecular beam epitaxy, other suitable epitaxial growth processes, or combinations thereof. The bottom S/D layer **252** reserves a space for a subsequently-formed backside S/D contact. In the illustrated embodiment, the bottom S/D layer **252** fills a bottom portion of the source trench **246** and is below the bottommost epitaxial layer **206**. The etch mask **250** is subsequently removed, for example, by a resist stripping process or other suitable process.

**[0039]** At operation **118**, the method **100** (FIG. 1B) forms S/D features **260** in the S/D trenches **246** in the region I. The resultant structure is shown in FIGS. 11A and 11B according to an embodiment. In some embodiments, the operation **118** may include epitaxially growing semiconductor materials such as epitaxially grown silicon, germanium, or silicon germanium. The S/D features **260** can be formed by any epitaxy processes including chemical vapor deposition (CVD) techniques (for example, vapor phase epitaxy and/or Ultra-High Vacuum CVD), molecular beam epitaxy, other suitable epitaxial growth processes, or combinations thereof. The S/D features **260** may be doped with n-type dopants and/or p-type dopants. In some embodiments, for n-type transistors, the S/D features **260** include silicon and can be doped with carbon, phosphorous, arsenic, other n-type dopant, or combinations thereof (for example, forming Si:C epitaxial S/D features, Si:P epitaxial S/D features, or Si:C:P epitaxial S/D features). In some embodiments, for p-type transistors, the S/D features **260** include silicon germanium or germanium, and can be doped with boron, other p-type dopant, or combinations thereof (for example, forming Si:Ge:B epitaxial S/D features). The S/D features **260** may include multiple epitaxial semiconductor layers having different levels of dopant density. In some embodiments, annealing processes (e.g., rapid thermal annealing (RTA) and/or laser annealing) are performed to activate dopants in the S/D features **260**.

**[0040]** At operation **120**, the method **100** (FIG. 1B) forms implanted P+ region and N+ region in the fin **210b** in the region II to form ESD devices. The resultant structure is shown in FIGS. 12A and 12B according to an embodiment. The operation **120** may include performing an n-type impurity implantation (or doping) to form a heavily doped n-type region **262** (or N+ region **262**) and performing a p-type impurity implantation to form a heavily doped p-type region **264** (or P+ region **264**). Throughout the description, the term “heavily doped” indicates impurity concentrations of greater than about  $10^{20}/\text{cm}^3$ . However, it is appreciated that the term “heavily doped” is a term of art, and is related to the specific technology generation used for forming the integrated circuits of the embodiments. When the conductivity type is p-type, suitable impurities include boron, aluminum, or other suitable p-type dopants. When the conductivity type is n-type, suitable impurities include phosphorous, arsenic, or other suitable n-type dopants. In the illustrated embodiment, the implantations are bounded in regions between the adjacent dummy gate stacks **232** in the region II. As is known in the art, masks, such as resists, may be formed to cover regions not to be implanted. Each of the P+ region **264** and the N+ region **262** may have a width along the Y direction substantially equal to the width **W2** of the fin **210b** and a length **L4** along the X direction ranging from about 30 nm to about 200 nm, in some embodiments. In the illustrated



embodiment, a doping depth of the P+ region **264** and the N+ region **262** extends beyond the bottommost epitaxial layer **206** and into the top substrate portion **202a** for a distance *d*. In one embodiment, the distance *d* ranges from about 0 nm to about 30 nm. Such a distance *d* improves RC performance of the ESD devices in the embodiment. A middle portion of the fin **210b** may remain to be un-implanted. The un-implanted middle portion of the fin **210b** is between and adjoining the N+ region **262** and the P+ region **264**. Referring to FIG. **12B** and also FIG. **16A** for a top view, the un-implanted middle portion may have a length *L5* along the X direction ranging from about 30 nm to about 500 nm, in some embodiments. In various embodiments, a ratio of *L5/L4* ranges from about 0.2 to about 2.5. If the ratio is less than 0.2, the relative small distance between P+/N+ regions may cause dopants diffuse into each other and reduce effective implant region widths. If the ratio is larger than 2.5, the relatively large distance between P+/N+ regions may increase resistance along ESD paths and degrade ESD performance. In the illustrated embodiment, the dummy gate stacks **232** is laterally distant from the P+/N+ regions for a gap *G2* (measured from sidewalls of the gate spacers **242**) ranging from about 10 nm to about 60 nm. Alternatively, the dummy gate stacks **232** may abut with the P+/N+ regions (*G2*=0) or even partially overlap with edges of the P+/N+ regions, respectively.

[0041] In the resulting structure after operation **120**, a P-N junction is formed between the P+ region **264** and the N+ region **262**. The P-N junction is along the lengthwise direction of the fin **210b**. The resulting structure is an ESD diode in the illustrated embodiment. By having the P+/N+ regions in the same fin, instead of crossing multiple fins, the ESD current conductivity is enhanced and ESD parasitic capacitance is reduced. By having the dummy gate stacks and accordingly the subsequently-formed metal gate stacks outside of the P+/N+ regions, the ESD parasitic capacitance between ESD diodes and gate structures is further reduced. The relatively large width of the fin **210b** also provides a low resistivity ESD current path that enhances the ESD device performance.

[0042] At operation **122**, the method **100** (FIG. **1B**) forms an inter-layer dielectric (ILD) layer **268** on sidewalls of the dummy gate stacks **232** and covering the fins **210**. Referring to FIGS. **13A** and **13B**, in some embodiments of operation **122**, a contact etch stop layer (CESL) **266** is also formed prior to forming the ILD layer **268**. In some examples, the CESL **266** includes a silicon nitride layer, silicon oxide layer, a silicon oxynitride layer, and/or other materials known in the art. The CESL **266** may be formed by plasma-enhanced chemical vapor deposition (PECVD) process and/or other suitable deposition or oxidation processes. In some embodiments, the ILD layer **268** includes materials such as tetraethylorthosilicate (TEOS) oxide, un-doped silicate glass, or doped silicon oxide such as borophosphosilicate glass (BPSG), fused silica glass (FSG), phosphosilicate glass (PSG), boron doped silicon glass (BSG), and/or other suitable dielectric materials. The ILD layer **268** may be deposited by a PECVD process or other suitable deposition technique. In some embodiments, after formation of the ILD layer **268**, the device **200** may be subject to a high thermal budget process to anneal the ILD layer. In some examples, after depositing the ILD layer **268**, a planarization process may be performed to remove excessive dielectric materials. For example, a planarization process includes a CMP pro-

cess which removes portions of the CESL **266** and the ILD layer **268** overlying the dummy gate stacks **232** and planarizes a top surface of the device **200**. In some embodiments, the CMP process exposes the dummy electrode layer **236**.

[0043] At operation **124**, the method **100** (FIG. **1B**) removes the dummy gate stack **232** to form gate trenches **270** in both the regions I and II, as shown in FIGS. **13A** and **13B**. A final gate structure (e.g., including a high-K dielectric layer and metal gate electrode) may be subsequently formed in the gate trench **270**, as will be described below. Operation **124** may include one or more etching processes that are selective to the materials in the dummy dielectric layer **234** and the dummy electrode layer **236** of the dummy gate stacks **232**. For example, the removal of the dummy gate stacks **232** may be performed using a selective etch process such as a selective wet etch, a selective dry etch, or a combination thereof. The epitaxial layers **206** and **208** of the fins **210a** in the region I and the topmost epitaxial layer **208** of the fin **210b** in the region II are exposed in the gate trenches **270**.

[0044] At operation **126**, the method **100** (FIG. **1B**) removes the epitaxial layers **206** from the fins **210a** exposed in the gate trenches **270** in the region I. The resultant structure is shown in FIGS. **13A** and **13B** according to an embodiment. In an embodiment, the epitaxial layers **206** include SiGe and the epitaxial layers **208** include Si, allowing for the selective removal of the epitaxial layers **206**. In an embodiment, the epitaxial layers **206** are removed by a selective wet etching process. In some embodiments, the selective wet etching includes an APM etch (e.g., ammonia hydroxide-hydrogen peroxide-water mixture). In some embodiments, the selective removal includes SiGe oxidation followed by a SiGeOx removal. For example, the oxidation may be provided by O<sub>3</sub> clean and then SiGeOx removed by an etchant such as NH<sub>4</sub>OH. In the region II, the topmost epitaxial layer **208** serves as an etch stop and protects the underneath epitaxial layers **206** from etching.

[0045] The method **100** then proceeds to operation **128** (FIG. **1B**) where gate structures are formed. The resultant structure is shown in FIGS. **14A** and **14B** according to an embodiment. The gate structures may be high-K/metal gate (HK MG) stacks, however other compositions are possible. In some embodiments, the gate structures form the gate associated with the multi-channels provided by the plurality of channel layers (e.g., nanosheets or nanowires having gaps therebetween) in the channel regions in the region I. In an embodiment of operation **128**, HK MG stacks **274** is formed within the gate trenches **270** of the device **200**. In various embodiments, the HK MG stacks **274** include an interfacial layer (not shown), a high-K gate dielectric layer **276** formed over the interfacial layer, and a gate electrode layer **278** formed over the high-K gate dielectric layer **276**. High-K gate dielectrics, as used and described herein, include dielectric materials having a high dielectric constant, for example, greater than that of thermal silicon oxide (~3.9). The gate electrode layer used within HK MG stack may include a metal, metal alloy, or metal silicide. Additionally, the formation of the HK MG stack may include depositions to form various gate materials, one or more liner layers, and one or more CMP processes to remove excessive gate materials and thereby planarize a top surface of the semiconductor device **200**. Interposing the HK MG stacks **274** and the S/D features **260** are the inner spacers **248**, providing isolation.

[0046] In some embodiments, the interfacial layer of the HK MG stacks **274** may include a dielectric material such as silicon oxide (SiO<sub>2</sub>), HfSiO, or silicon oxynitride (SiON). The interfacial layer may be formed by chemical oxidation, thermal oxidation, atomic layer deposition (ALD), chemical vapor deposition (CVD), and/or other suitable method. The high-K gate dielectric layer **276** of the HK MG stack **274** may include a high-K dielectric such as hafnium oxide (HfO<sub>2</sub>). Alternatively, the high-K gate dielectric layer **276** of the HK MG stack **274** may include other high-K dielectrics, such as TiO<sub>2</sub>, HfZrO, Ta<sub>2</sub>O<sub>3</sub>, HfSiO<sub>4</sub>, ZrO<sub>2</sub>, ZrSiO<sub>2</sub>, LaO, AlO, ZrO, TiO, Ta<sub>2</sub>O<sub>5</sub>, Y<sub>2</sub>O<sub>3</sub>, SrTiO<sub>3</sub> (STO), BaTiO<sub>3</sub> (BTO), BaZrO, HfZrO, HfLaO, HfSiO, LaSiO, AlSiO, HfTaO, HfTiO, (Ba,Sr)TiO<sub>3</sub> (BST), Al<sub>2</sub>O<sub>3</sub>, Si<sub>3</sub>N<sub>4</sub>, oxynitrides (SiON), combinations thereof, or other suitable material. The high-K gate dielectric layer **276** may be formed by ALD, physical vapor deposition (PVD), CVD, oxidation, and/or other suitable methods.

[0047] The gate electrode layer **278** of the HK MG stacks **274** may include a single layer or alternatively a multi-layer structure, such as various combinations of a metal layer with a selected work function to enhance the device performance (work function metal layer), a liner layer, a wetting layer, an adhesion layer, a metal alloy or a metal silicide. By way of example, the gate electrode layer **278** of HK MG stacks **274** may include Ti, Ag, Al, TiAlN, TaC, TaCN, TaSiN, Mn, Zr, TiN, TaN, Ru, Mo, WN, Cu, W, Re, Ir, Co, Ni, other suitable metal materials or a combination thereof. In various embodiments, the gate electrode layer **278** of the HK MG stacks **274** may be formed by ALD, PVD, CVD, e-beam evaporation, or other suitable process. Further, the gate electrode layer **278** may be formed separately for N-FET and P-FET transistors which may use different metal layers (e.g., for providing an N-type or P-type work function). In various embodiments, a CMP process may be performed to remove excessive metal from the gate electrode layer **278** of the HK MG stacks **274**, and thereby provide a substantially planar top surface of the HK MG stacks **274**. The HK MG stacks **274** includes portions that interpose each of the epitaxial layers (channel layers) **208**, which form channels of the multi-gate transistors in the region I.

[0048] At operation **130**, the method **100** (FIG. 1B) forms various features on the frontside of the device **200**. For example, the operation **130** may form contact openings exposing underneath N+/P+ regions **262/264** by etching the ILD layer **268** and the CESL **266**, contacts **282a** landing on S/D features **260** in the region I and contacts **282b** landing on N+/P+ regions **262/264** in the region II by filling the contact openings with conductive materials, as well as various other contacts/vias/lines and multilayers interconnect features (e.g., metal layers and interlayer dielectrics) on the frontside of the device **200** (denoted as metal wiring layer **284**). The resultant structure is shown in FIGS. **15A** and **15B** according to an embodiment. In furtherance of the example, the metal wiring layer **284** may include a multi-layer interconnection, such as vertical interconnects (e.g., vias or contacts) and horizontal interconnects (e.g., metal lines). The various interconnection features may employ various conductive materials including copper, tungsten, and/or silicide. In one example, a damascene and/or dual damascene process is used to form a copper related multi-layer interconnection structure.

[0049] Referring to FIG. **16A** depicting a top view of the device **200** after the operation **130** according to an embodi-

ment, the contacts **282b** in the region II are spaced along the X direction and extend along the Y direction. Each of the contacts **282b** may have a width along the X direction ranging from about 10 nm to about 80 nm and a length along the Y direction ranging from about 40 nm to about 950 nm. The contacts **282b** may be deposited in a portion of the P+/N+ regions **264/262** that has an edge-to-edge distance L6' to inside edges (i.e., opposing edges) of the P+/N+ regions and an edge-to-edge distance L6 to outside edges (i.e., edges facing HK MG stacks **274**) of the P+/N+ regions, respectively. In various embodiments, L6' is less than L6. In other words, the contacts **282b** are closer to the inside edges of the P+/N+ regions than the outside edges. In some embodiment, a ratio of L6'/L6 is less than about 0.4. For example, the distance L6' may be less than about 10 nm and the distance L6 may range between about 30 nm to about 60 nm. Referring to FIG. **16B** depicting a top view of the device **200** after the operation **130** according to yet another embodiment as discussed above in association with FIG. **5B**, the gate stacks **274** are deposited on the STI features **220** and the P+/N+ regions extends to lateral ends of the fin **210b**. A distance G1 between two opposing edges of the gate stacks **274** and the fin **210b** is also a distance G2 between two opposing edges of the gate stacks **274** and the P+/N+ regions, respectively, in a range from about 10 nm to about 60 nm. By having the gate stacks **274** outside of the fin **210b**, the ESD parasitic capacitance between ESD diodes and gate structures is further reduced.

[0050] At operation **132**, the method **100** (FIG. 1B) flips the device **200** upside down and attaches the frontside of the device **200** to a carrier **286**, such as shown in FIGS. **17A** and **17B**. This makes the device **200** accessible from the backside of the device **200** for further processing. The operation **132** may use any suitable attaching processes, such as direct bonding, hybrid bonding, using adhesive, or other bonding methods. The operation **132** may further include alignment, annealing, and/or other processes. The carrier **286** may be a silicon wafer in some embodiments. In FIGS. **2-23B**, the “z” direction points from the backside of the device **200** to the frontside of the device **200**, while the “-z” direction points from the frontside of the device **200** to the backside of the device **200**.

[0051] At operation **134**, the method **100** (FIG. 1C) thins down the device **200** from the backside of the device **200** until the bottom S/D layer **252** is exposed from the backside of the device **200**. The resultant structure is shown in FIGS. **18A** and **18B** according to an embodiment. The thinning process may include a mechanical grinding process and/or a chemical thinning process. A substantial amount of substrate material may be first removed from the substrate **202** during a mechanical grinding process. The dielectric capping layer **203** may serve as a CMP stop layer during the removal of the bottom substrate portion **202b**. Afterwards, a chemical thinning process may apply an etching chemical to the backside of the substrate **202** to remove the dielectric capping layer **203** and further thin down the top substrate portion **202a** until the bottom S/D layer **252** is exposed.

[0052] At operation **136**, the method **100** (FIG. 1C) removes the top substrate portion **202a** in the region I from the backside of the device **200**. The resultant structure is shown in FIGS. **19A** and **19B** according to an embodiment. The operation **136** may include first forming an etch mask (not shown) covering the region II. The operation **136** then applies an etching process that is tuned to be selective to the

materials of the semiconductor material (e.g. silicon) in the upper substrate portion **202a** and with no (or minimal) etching to the bottom S/D layer **252**. The etching process can be dry etching, wet etching, reactive ion etching, or other etching methods. After the removing of the top substrate portion from the region I, the metal gate stacks **274**, the inner spacers **248**, and the S/D features **260** are exposed at the backside of the device **200**. The etch mask is subsequently removed, for example, by a resist stripping process or other suitable process. The top substrate portion **202a** remains in the region II of the device **200**.

**[0053]** At operation **138**, the method **100** (FIG. 1C) deposits a dielectric layer **288** on the backside of the device **200**. The resultant structure is shown in FIGS. **20A** and **20B** according to an embodiment. The operation **138** may first include a chemical thinning process to recess the top substrate portion **202a** in the region II to a level lower than the bottom S/D layer **252** in the region I. The operation **138** then deposits the dielectric layer **288** covering both the regions I and II. The dielectric layer **288** may comprise tetraethyl-orthosilicate (TEOS) oxide, un-doped silicate glass, or doped silicon oxide such as borophosphosilicate glass (BPSG), fluoride-doped silica glass (FSG), phosphosilicate glass (PSG), boron doped silicon glass (BSG), a low-k dielectric material, other suitable dielectric material, or combinations thereof. The dielectric layer **288** may be formed by PE-CVD (plasma enhanced CVD), F-CVD (flowable CVD), or other suitable methods. The operation **138** may also include a CMP process to planarize the dielectric layer **288** to remove excessive dielectric materials from the backside of the device **200**. Since the dielectric layer **288** and the bottom S/D layer **252** include different materials so that the bottom S/D layer **252** may act as a CMP stop when the dielectric layer **288** is planarized by the CMP process. After the operation **138**, the bottom S/D layer **252** is exposed in the region I, while the top substrate portion **202a** remains covered by the dielectric layer **288** in the region II.

**[0054]** At operation **140**, the method **100** (FIG. 1C) selectively etches the bottom S/D layer **252** to form a backside trench **290** surrounded by the dielectric layer **288**. The backside trench **290** expose a surface of the source feature **260** from the backside. The resultant structure is shown in FIGS. **21A** and **21B** according to an embodiment. In the present embodiment, the operation **140** applies an etching process that is tuned to be selective to the materials of the material (e.g. SiGe) in the bottom S/D layer **252** and with no (or minimal) etching to the dielectric layer **288**. In the present embodiment, the etching process also etches the source feature **260** to recess it to a level that is below drain features **260** that are still covered by the dielectric layer **288**. In furtherance of the embodiment, the source feature **260** is recessed to be below the dielectric layer **288** and the bottommost inner spacer **248**. The operation **140** may apply more than one etching processes. For example, it may apply a first etching process to selectively remove the bottom S/D layer **252**, and then apply a second etching process to selectively recess the source feature **260** to the desired level, where the first and the second etching processes use different etching parameters such as using different etchants. The etching process(es) can be dry etching, wet etching, reactive ion etching, or other etching methods.

**[0055]** At operation **142**, the method **100** (FIG. 1C) deposits a dielectric liner **292** on sidewalls of the backside trench

**290**, as shown in FIGS. **22A** and **22B** according to an embodiment. In some embodiments, the dielectric liner **292** is conformally deposited to have a substantially uniform thickness along the various surfaces of the dielectric layer **288**, the inner spacers **248**, and the recessed source feature **260**. In various embodiment, the dielectric liner **292** may include  $\text{La}_2\text{O}_3$ ,  $\text{Al}_2\text{O}_3$ , SiOCN, SiOC, SiCN,  $\text{SiO}_2$ , SiC, ZnO, ZrN,  $\text{Zr}_2\text{Al}_3\text{O}_9$ ,  $\text{TiO}_2$ ,  $\text{TaO}_2$ ,  $\text{ZrO}_2$ ,  $\text{HfO}_2$ ,  $\text{Si}_3\text{N}_4$ ,  $\text{Y}_2\text{O}_3$ , AlON, TaCN, ZrSi, combinations thereof, or other suitable material(s). The dielectric liner **292** may be deposited using ALD, CVD, or other suitable methods, and may have a thickness of about 0.5 nm to about 10 nm in various embodiments. The operation **142** further includes a breakthrough etching process for breaking through, and removing the majority of, the horizontal portions of the dielectric liner **292**. In some embodiments, the breakthrough etching process may include an anisotropic dry etch process, or the like. In some embodiments where the dielectric liner **292** is formed of an oxide compound, the BT etching process is a reactive ion etch (RIE) process with etch process gases including  $\text{CHF}_3$ , Ar,  $\text{CF}_4$ ,  $\text{N}_2$ ,  $\text{O}_2$ ,  $\text{CH}_2\text{F}_2$ ,  $\text{SF}_3$ , the like, or a combination thereof. In the illustrated embodiment, as a result of the operation **142**, portions of the dielectric liner **292** remain on sidewalls of the backside trench **290**, including surfaces of the dielectric layer **288** and the inner spacers **248**. The recessed source feature **260** is exposed again in the backside trench **290** after the breakthrough etching process.

**[0056]** At operation **144**, the method **100** (FIG. 1C) forms a silicide feature **294** in the backside trench **290** over the source feature **260**, as shown in FIGS. **22A** and **22B** according to an embodiment. In an embodiment, the operation **144** first deposits one or more metals into the backside trench **290**, performing an annealing process to the device **200** to cause reaction between the one or more metals and the source feature **260** to produce the silicide feature **294**, and removing un-reacted portions of the one or more metals, leaving the silicide feature **294** in the backside trench **290**. The one or more metals may include titanium (Ti), tantalum (Ta), tungsten (W), nickel (Ni), platinum (Pt), ytterbium (Yb), iridium (Ir), erbium (Er), cobalt (Co), or a combination thereof (e.g., an alloy of two or more metals) and may be deposited using CVD, PVD, ALD, or other suitable methods. The silicide features **294** may include titanium silicide (TiSi), nickel silicide (NiSi), tungsten silicide (WSi), nickel-platinum silicide (NiPtSi), nickel-platinum-germanium silicide (NiPtGeSi), nickel-germanium silicide (NiGeSi), ytterbium silicide (YbSi), platinum silicide (PtSi), iridium silicide (IrSi), erbium silicide (ErSi), cobalt silicide (CoSi), a combination thereof, or other suitable compounds.

**[0057]** At operation **146**, the method **100** (FIG. 1C) forms backside S/D contact **296** in the backside trench **290** over the silicide feature **294** in the region I. The resultant structure is shown in FIGS. **22A** and **22B** according to an embodiment. In an embodiment, the backside S/D contact **296** may include a conductive barrier layer and a metal fill layer over the conductive barrier layer. The conductive barrier layer may include titanium (Ti), tantalum (Ta), tungsten (W), cobalt (Co), ruthenium (Ru), or a conductive nitride such as titanium nitride (TiN), titanium aluminum nitride (TiAlN), tungsten nitride (WN), tantalum nitride (TaN), or combinations thereof, and may be formed by CVD, PVD, ALD, and/or other suitable processes. The metal fill layer may include tungsten (W), cobalt (Co), molybdenum (Mo), ruthenium (Ru), nickel (Ni), copper (Cu), or other metals,

and may be formed by CVD, PVD, ALD, plating, or other suitable processes. In some embodiments, the conductive barrier layer is omitted in the backside S/D contact **296**. The dielectric liner **292** serves as a dielectric barrier layer preventing the metallic material in the backside S/D contact **296** from diffusing into the dielectric layer **288**.

**[0058]** At operation **148**, the method **100** (FIG. 1C) performs further fabrication processes to the device **200**. For example, it may form a metal wiring layer **298**, such as backside power rails, on the backside of the device **200**. The resultant structure is shown in FIGS. **23A** and **23B** according to an embodiment. The metal wiring layer **298** electrically connects to the backside S/D contact **296** in the region I. In an embodiment, the metal wiring layer may be formed using a damascene process, a dual-damascene process, a metal patterning process, or other suitable processes. The metal wiring layer may include tungsten (W), cobalt (Co), molybdenum (Mo), ruthenium (Ru), copper (Cu), nickel (Ni), titanium (Ti), tantalum (Ta), titanium nitride (TiN), tantalum nitride (Ta<sub>3</sub>N<sub>5</sub>), or other metals, and may be deposited by CVD, PVD, ALD, plating, or other suitable processes. Having a backside metal wiring layer beneficially increases the number of metal tracks available in the device **200** for directly connecting to source/drain contacts and vias. It also increases the gate density for greater device integration than other structures without the backside metal wiring layer. In some embodiments, one of the two contacts **282b** of the ESD diode (e.g., either the one disposed on the P+ region **264** or the one disposed on the N+ region **262**) is electrically coupled to the metal wiring layer **284** on the frontside of the device **200**, while the other contact **282b** is electrically coupled to the metal wiring layer **298** on the backside of the device **200** such as by a through-substrate-via (TSV). The method **100** at operation **148** may also form one or more interconnect layers on the backside of the device **200**, form passivation layers on the backside of the device **200**, perform other BEOL processes, and remove the carrier **286**. As shown in FIGS. **23A** and **23B**, even though bulk semiconductor substrate is removed from the region I in the illustrated multi-gate process flow as part of the backside metal wiring layer formation, the top substrate portion **202a** is preserved in the region II under the epitaxial layers **206/208** and the N+/P+ regions **262/264**, which provides extra ESD current path in the substrate and thus less ESD resistance.

**[0059]** Although not intended to be limiting, embodiments of the present disclosure provide one or more of the following advantages. For example, embodiments of the present disclosure form ESD protection device and methods of fabrication thereof that is compatible with multi-gate transistor process flows. In various embodiment, the methods in the present disclosure preserves a bulk semiconductor substrate for the ESD protection device after a backside substrate thinning process, which provides more current path for electrostatic discharging events. Further, some embodiments of the present disclosure form implant regions on the same fin element comprising a stack of epitaxial layers, which enhances ESD current conductivity and reduces ESD parasitic capacitance. Embodiments of the process flow also supports the formation of multi-gate transistors with backside metal wiring layer, such as backside power rails, to increase the number of metal tracks available in an integrated circuit and increase the gate density for greater device

integration. Embodiments of the present disclosure can be readily integrated into existing semiconductor manufacturing processes.

**[0060]** In one example aspect, the present disclosure is directed to a method of manufacturing a semiconductor device. The method includes providing a structure having a frontside and a backside, the structure including a substrate and a stack of a first type and a second type epitaxial layers having different material compositions alternatively stacked above the substrate, wherein the stack is at the frontside of the structure and the substrate is at the backside of the structure; patterning the stack, thereby forming a fin above the substrate; implanting a first dopant into a first region of the fin, the first dopant having a first conductivity type; implanting a second dopant into a second region of the fin, the second dopant having a second conductivity type opposite the first conductivity type; and forming a first contact on the first region and a second contact on the second region. In some embodiments, the first and second regions are separated by a middle portion of the fin. In some embodiments, the method also includes forming first and second dummy gate stacks on the fin; depositing an inter-layer dielectric (ILD) layer on sidewalls of the first and second dummy gate stacks and covering the fin; and replacing the first and second dummy gate stacks with first and second metal gate stacks. In some embodiments, the first and second contacts are between the first and second metal gate stacks. In some embodiments, each of the first and second metal gate stacks has a width less than that of the fin. In some embodiments, the method also includes thinning down the substrate from the backside of the structure; and depositing a dielectric layer at the backside of the structure. In some embodiments, the method further includes forming an isolation structure surrounding the fin; and forming at least a gate stack directly above the isolation structure. In some embodiments, where the fin is a first fin and the patterning of the stack also forms a second fin, the method further includes removing the second type epitaxial layers from the second fin, thereby exposing the first type epitaxial layers; and forming a metal gate stack engaging the first type epitaxial layers in the second fin. In some embodiments, the second fin has a width less than that of the first fin.

**[0061]** In another example aspect, the present disclosure is directed to a method of manufacturing a semiconductor device. The method includes forming a fin extruding from a substrate, wherein the fin includes a first type and a second type epitaxial layers on a semiconductor substrate, the first type and the second type epitaxial layers having different material compositions and alternatingly disposed in a vertical direction; forming first and second gate structures above the fin; doping a first region of the fin with a first dopant of a first conductivity type; doping a second region of the fin with a second dopant of a second conductivity type opposite the first conductivity type, wherein the first and second regions are between the first and second gate structures; forming a dielectric layer on sidewalls of the first and second gate structures; forming contact holes in the dielectric layer exposing the first and second regions; and forming conductive contacts in the contact holes. In some embodiments, the doping of the first and second regions has a doping depth extending into the semiconductor substrate. In some embodiments, the method further includes thinning down the semiconductor substrate; and depositing an oxide layer under the semiconductor substrate. In some embodi-

ments, the fin has a width larger than that of either the first or second gate structure. In some embodiments, after the doping of the first and second regions of the fin, a middle portion of the fin is substantially free of the first and second dopants.

**[0062]** In yet another example aspect, the present disclosure is directed to an electrostatic discharging (ESD) device. The ESD device includes a semiconductor substrate; a stack of a first type and a second type epitaxial layers on the semiconductor substrate, the first type and the second type epitaxial layers having different material compositions and the first type and the second type epitaxial layers being alternately disposed in a vertical direction; a first implant region in the stack, wherein the first implant region has a first conductivity type; and a second implant region in the stack, wherein the second implant region has a second conductivity type opposite the first conductivity type. In some embodiments, the ESD device further includes a middle region in the stack between and adjoining the first and second implant regions. In some embodiments, the ESD device further includes an oxide layer under the semiconductor substrate. In some embodiments, the ESD device further includes first and second gate stacks above the stack, wherein the first and second implant regions are between the first and second gate stacks. In some embodiments, the first and second gate stacks are directly above the stack. In some embodiments, the ESD device further includes an isolation structure above the semiconductor substrate and surrounding the stack, where the first and second gate stacks are directly above the isolation structure.

**[0063]** The foregoing outlines features of several embodiments so that those of ordinary skill in the art may better understand the aspects of the present disclosure. Those of ordinary skill in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those of ordinary skill in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A semiconductor device, comprising:
  - a stack disposed in a first region of the semiconductor device, the stack having first type epitaxial layers and second type epitaxial layers alternately arranged in a vertical direction, the first type epitaxial layers having a first material composition and the second type epitaxial layers having a second material composition different from the first material composition, the stack having a first implant region of a first conductivity type and a second implant region of a second conductivity type opposite the first conductivity type;
  - a plurality of channel members disposed in a second region of the semiconductor device and stacked in the vertical direction, the channel members having the first material composition; and
  - a metal gate wrapping around at least one of the channel members.
2. The semiconductor device of claim 1, further comprising:

first and second conductive structures disposed in the first region, wherein the first and second implant regions are laterally between the first and second conductive structures.

3. The semiconductor device of claim 2, wherein the first and second conductive structures are disposed on a top surface of the stack.

4. The semiconductor device of claim 2, wherein the stack is disposed between the first and second conductive structures.

5. The semiconductor device of claim 1, further comprising:

a semiconductor substrate disposed under the stack, wherein at least one of the first and second implant regions extends downwardly through a bottommost one of the first and second type epitaxial layers and into a top portion of the semiconductor substrate.

6. The semiconductor device of claim 5, wherein the semiconductor substrate does not laterally extend into the second region.

7. The semiconductor device of claim 1, further comprising:

a first contact electrically coupled to the first implant region; and

a second contact electrically coupled to the second implant region.

8. The semiconductor device of claim 7, further comprising:

a frontside dielectric layer disposed on the stack, wherein each of the first and second contact extends through the dielectric layer.

9. The semiconductor device of claim 1, further comprising:

a backside dielectric layer underneath the stack and the channel members, wherein the backside dielectric layer extends continuously from the first region to the second region.

10. The semiconductor device of claim 1, wherein the first implant region is laterally distant from the second implant region.

11. A semiconductor device, comprising:

a semiconductor substrate;

an epitaxial stack disposed above the semiconductor substrate, the epitaxial stack including first and second type epitaxial layers, the first and second type epitaxial layers having different material compositions, and the first and second type epitaxial layers being alternately disposed in a vertical direction;

a first implant region in the epitaxial stack, the first implant region having a first dopant of a first conductivity type;

a second implant region in the epitaxial stack, the second implant region having a second dopant of a second conductivity type opposite the first conductivity type; and

first and second conductive structures disposed above the epitaxial stack,

wherein the first and second implant regions are between the first and second conductive structures.

12. The semiconductor device of claim 11, wherein in a top view of the semiconductor device the first and second conductive structures are within a boundary of a top surface of the epitaxial stack.

**13.** The semiconductor device of claim **11**, wherein in a top view of the semiconductor device the epitaxial stack is positioned between the first and second conductive structures.

**14.** The semiconductor device of claim **11**, wherein each of the first and second implant regions extends downwardly through the epitaxial stack and into a top portion of the semiconductor substrate.

**15.** The semiconductor device of claim **11**, further comprising:

a backside dielectric layer disposed under the semiconductor substrate.

**16.** The semiconductor device of claim **11**, further comprising:

a first contact feature electrically coupled to the first implant region; and

a second contact feature electrically coupled to the second implant region, wherein the first and second contact features are laterally between the first and second conductive structures.

**17.** A method of manufacturing a semiconductor device, comprising:

providing a structure having a substrate and a stack protruding from the substrate, the stack having first semiconductor layers and second semiconductor layers alternately arranged in a vertical direction, the first and second semiconductor layers having different material compositions;

implanting a first dopant into a first region of the stack, the first dopant having a first conductivity type;

implanting a second dopant into a second region of the stack, the second dopant having a second conductivity type opposite the first conductivity type;

depositing a dielectric layer over the stack;

forming a first contact through the dielectric layer and electrically coupled to the first region; and

forming a second contact through the dielectric layer and electrically coupled to the second region.

**18.** The method of claim **17**, further comprising:

forming a first conductive structure above the first region; and

forming a second conductive structure above the second region,

wherein the dielectric layer extends from a sidewall of the first conductive structure to a sidewall of the second conductive structure.

**19.** The method of claim **18**, wherein in a top view of the semiconductor device the first and second conductive structures are fully within a contour of a top surface of the stack.

**20.** The method of claim **17**, wherein the stack includes a third region substantially free of the first and second dopants, and the third region is laterally between the first and second regions of the stack.

\* \* \* \* \*